

Entropy Collapse Has a Direction: Correcting the Visitation Channel in Token-Level Entropy Control for RLVR

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Abstract

Reinforcement learning with verifiable rewards (RLVR) reliably drives a policy’s entropy toward zero, and a growing family of methods intervenes by re-weighting the policy loss with a per-token estimate of how much each token’s update costs in entropy. We show that every such method estimates only half of the relevant quantity. The gradient of the *trajectory* entropy splits exactly into two terms: a *local* channel, which asks how the conditional distribution at a visited state flattens or sharpens, and a *visitation* channel, which asks how the probability of *reaching* that state changes. Existing token-level estimators capture the first and are structurally blind to the second. We show the blindness is worse than an omission: the local estimator is exactly zero at a uniform branch point, that zero is the correct first-order answer, and the visitation term’s prefactor is simultaneously maximal there — the two channels are *anti-correlated*, so no refinement of the local term can recover the missing signal. We then derive the missing channel into a per-token advantage A_H . Its reward-to-go is the remaining conditional entropy H_{togo} , and the sibling-prefix baseline that keeps the estimator unbiased is supplied by a tree-structured rollout in which several samples share a prefix and diverge at scheduled depths. A_H sums to zero within a sibling set by construction, so the correction cannot change the average amount of damping — it can only decide which of two siblings is damped. A five-arm design, which runs the full pipeline twice with A_H ’s value destroyed in two different ways, measures that consequence: switching the damping apparatus on or off moves the aggregate entropy ($t = -5.58$ and $+18.52$), while changing what A_H carries with the apparatus held fixed does not move it detectably ($t = -1.42$), and accuracy separates the two groups the other way round. The aggregate and the correction are dissociated, as the construction says they must be. We forecast H_{togo} with parallel multi-token-

prediction heads, state the four approximations that estimation requires and what each discards, and report our own measurement that the forecast comes out close to a scalar multiple of a quantity the trainer already computes — so the forecaster is one estimator of this reward-to-go, not a part the decomposition depends on.

1 Introduction

Policy-gradient fine-tuning with verifiable rewards sharpens a language model’s output distribution. Measured as the mean per-token entropy of the policy, this sharpening is fast, monotone, and, past a point, coincident with the end of useful learning. The standard reading is that the collapse is the problem, and the standard remedy is to spend part of the update budget slowing it down: an entropy bonus, a KL anchor, a clip on the tokens that lose the most entropy, or — in the most recent line of work — a per-token re-weighting of the policy loss by a first-order estimate of the entropy each token’s update destroys.

We argue that the accounting behind that remedy is incomplete, and that the incompleteness is structural rather than a matter of estimator quality.

Write $\mathcal{H}(\pi_\theta \mid x)$ for the entropy of the distribution the policy induces over whole responses to a prompt x . The autoregressive factorization turns it into an expectation of a sum of per-token conditional entropies, and the gradient of that expression splits, by the product rule, into exactly two terms. The first holds the visited states fixed and asks how the conditional distribution at each one changes: this is what a per-token entropy estimate measures. The second holds the per-state entropies fixed and asks how the probability of *visiting* each state changes. An update that concentrates mass on one continuation makes the states downstream of that continuation more likely and the states downstream of its abandoned siblings less likely; the diversity those siblings carried leaves the trajectory

distribution even though no conditional distribution anywhere became more peaked.

Our first contribution is to show that the local channel is not merely blind to the second term but *anti-correlated* with it. The local estimator factors into the advantage, a saturation factor, and the deviation of the sampled token’s surprisal from the mean surprisal — and that last factor has mean zero by the definition of entropy. At a branch point where several continuations are close to equiprobable it vanishes identically. The vanishing is not an estimator failure: a uniform distribution maximizes entropy, so the first-order change is genuinely zero and the damage is second order (Section 4.4). Meanwhile the visitation term’s prefactor is $1 - \pi_\theta(y_t | s_t)$, which is maximal precisely at those flat states. The local channel therefore spends its budget where the conditional entropy is fragile — rare tokens sampled from confident distributions — and spends nothing where the trajectory entropy is most at risk.

Our second contribution is an estimator for the missing channel that requires no additional rollouts. Eliminating the non-causal cross terms turns the visitation term into a REINFORCE estimator whose reward-to-go is the sum of the *remaining* conditional entropies, H_{togo} ; subtracting any function of the state leaves it unbiased. We estimate H_{togo} with parallel multi-token-prediction (MTP) heads read off the hidden state the policy already computes, and we obtain the state-only baseline from rollouts that share a prefix. Plain group-relative rollouts almost never share prefixes, so we schedule the sharing: a tree-structured sampler branches a single root at fixed depths, and the resulting sibling sets give the baseline an actual support (Section 9).

The resulting per-token score A_H is a deviation from a sibling mean and therefore sums to zero over each sibling set. This has a consequence we state up front because it determines what evidence can count: the method *cannot* raise the average weight at branch points, only spread the weights of siblings apart. “Total entropy went up” is therefore not a prediction of this method, and an experiment looking for it is testing a different hypothesis (Section 9.3).

We make the following contributions.

- An exact two-channel decomposition of the trajectory-entropy gradient, and an argument that the channel captured by current

token-level entropy-control methods is anti-correlated with the channel they miss (Sections 4–4.4).

- A derivation of the missing channel into a per-token advantage A_H with no approximation up to the point of estimating H_{togo} , together with an explicit ledger of the approximations that estimation requires, the sign of each one’s error, and the mechanism that absorbs it (Sections 5–10).
- A tree-structured rollout that supplies the sibling-prefix baseline, and a bound showing what fraction of positions can carry a non-zero A_H under *any* sampler (Section 9).
- A five-arm experiment that separates the contribution of A_H ’s *value* from that of the apparatus delivering it, by running the full pipeline twice with the value destroyed in two different ways, and the seed-paired contrasts it yields (Sections 11 and 12.4).
- The dissociation that construction predicts, measured rather than argued: the contrasts that switch the damping apparatus move the aggregate entropy ($t = -5.58$ and $+18.52$) while the contrasts that change only what A_H carries do not move it detectably ($t = -1.42$), and accuracy separates the same two groups in the opposite direction. The structural claim, not the effect size, is what this campaign establishes (Section 12.3).

2 Related Work

Entropy collapse in RLVR. Entropy decay under policy-gradient fine-tuning is well documented, and a range of interventions add an explicit entropy term to the objective or constrain the update so that high-entropy positions are preserved. Recent analyses report that naive entropy regularization does not prevent the collapse and that continuing to train under it degrades accuracy (Anonymous, 2025), which is consistent with our reading: the regularizer acts on the local channel, and the local channel is not where trajectory diversity is being lost.

Assumption. Both Hao et al. (2026) and this work rely on one structural assumption, which we restate so that our approximation ledger can refer to it.

Assumption 1 (parameter-independent softmax; Hao et al., 2026). For any state s , each action a carries an independent logit $z_{s,a}(\theta)$, so that a gradient step on the sampled token does not substantially move the logits of the other tokens at that state.

It is false for a network with shared parameters. We inherit it rather than introduce it, but our construction adds a second estimator that rests on it.

Token-level entropy control. The method we build on re-weights the clipped surrogate loss by a per-token first-order estimate of the entropy change that token’s update induces (Hao et al., 2026). That work observes that earlier interventions are “indirect” and that their effect is therefore limited and can fail. Our decomposition supplies a mechanism for that observation: the estimate is a first-order estimate of the local channel only, and it is exactly zero at the states where the visitation channel is largest.

Which tokens carry the gradient. Wang et al. (2025d) show that restricting policy-gradient updates to a small minority of high-entropy “forking” tokens matches full-gradient training. That result establishes that the identity of the tokens, not the aggregate entropy, is what matters, and it motivates a per-token treatment; it does not, however, distinguish the two channels, since forking tokens are identified by their local entropy. Our analysis predicts that a local criterion will select a systematically different set from the one the visitation channel would select.

Local stochasticity without global diversity. Wu et al. (2025a) report that token-level entropy can remain high while the diversity of final answers falls. This is precisely the signature our decomposition predicts when the visitation channel is unmanaged: per-state entropies are preserved — the local channel is doing its job — while the trajectory distribution concentrates.

Multi-token prediction. We use Medusa-style parallel prediction heads (Cai et al., 2024) not for their intended purpose (speculative decoding) but as an entropy forecaster: we need the entropy of the distribution over a token k steps ahead, not a good sample of that token. Section 6 shows that this choice of estimator has a signed, monotone error which we characterize and correct.

3 Preliminaries

Notation. A prompt is x , a response is $y = y_{1:T}$, and the state before emitting token t is $s_t = (x, y_{<t})$. We write $H(\pi_\theta(\cdot | s_t))$ for the entropy of the conditional distribution at a single position (lowercase, per-token) and $\mathcal{H}(\pi_\theta | x)$ for the entropy of the distribution over whole responses (uppercase, per-trajectory). The surprisal of a token a is $I_a := -\log \pi_\theta(a | s_t)$, so that $H = \mathbb{E}_{a \sim \pi_\theta} [I_a]$.

Group-relative policy optimization. We optimize with GRPO. For each prompt, n responses are sampled and scored, and each response receives the group-standardized advantage $A_i = (r_i - \text{mean}_j r_j) / \text{std}_j r_j$, broadcast to every token of that response. Replacing a learned value function with the group mean is what makes GRPO cheaper than PPO; it also means that a group whose members all receive the same reward contributes no gradient.

Token-level entropy control. Hao et al. (2026) leave the advantage untouched and instead multiply each token’s contribution to the clipped surrogate by a weight $\alpha_t \in [\lambda_{\min}, 1]$ derived from a first-order estimate of the entropy change that token’s update induces. Their Theorem 1 gives that estimate in closed form: under a parameter-independent softmax (their Assumption 1; our 2), the change in the conditional entropy at a *fixed* state s after one GRPO step is, to first order,

$$\Omega(s) = -\frac{\eta}{L} \mathbb{E}_a \left[\frac{\mathbb{I}_{\text{clip}}(s, a) A(s, a)}{\pi_{\text{old}}(a | s)} \cdot \pi_\theta(a | s) (1 - \pi_\theta(a | s)) \cdot (\log \pi_\theta(a | s) + H) \right], \quad (1)$$

where η is the learning rate, L the total decoded length in the update, and $\mathbb{I}_{\text{clip}}$ the PPO clipping indicator. The implementation evaluates the bracket at the sampled token rather than in expectation. The policy loss becomes $\sum_u \alpha_u \cdot \text{PPO-clip}(A_u, r_u)$, with

$$\alpha_u = \exp\left(-\alpha \frac{|\Omega_u|}{\max_v |\Omega_v|}\right), \quad \lambda_{\min} = e^{-\alpha}, \quad (2)$$

a batch-normalized exponential map into $[\lambda_{\min}, 1]$, so the weight can only attenuate. The factor $1/\pi_{\text{old}}$ in Eq. (1) will matter later: it makes Ω heavy-tailed, which is why the two channels must be normalized before they are combined (Section 8).

4 Two Channels

4.1 Step 1: the chain rule

The object we would like to protect is the entropy of the distribution the policy induces over *whole responses*,

$$\mathcal{H}(\pi_\theta | x) := - \sum_y \pi_\theta(y | x) \log \pi_\theta(y | x), \quad (3)$$

a sum over an exponentially large set. The autoregressive factorization $\pi_\theta(y | x) = \prod_t \pi_\theta(y_t | s_t)$ turns the logarithm into a sum and lets us exchange the order of summation:

$$\begin{aligned} \mathcal{H} &= - \sum_y \pi_\theta(y | x) \sum_{t=1}^T \log \pi_\theta(y_t | s_t) \\ &= \sum_{t=1}^T \mathbb{E}_{y \sim \pi_\theta} [-\log \pi_\theta(y_t | s_t)]. \end{aligned} \quad (4)$$

Each summand is still an expectation over whole responses, but the integrand depends on y only through $y_{\leq t}$. Conditioning on the prefix and applying the tower rule replaces the inner expectation with a per-position entropy,

$$\begin{aligned} \mathbb{E}_y [-\log \pi_\theta(y_t | s_t)] \\ = \mathbb{E}_{y_{\leq t}} \left[\underbrace{\mathbb{E}_{y_t \sim \pi_\theta(\cdot | s_t)} [-\log \pi_\theta(y_t | s_t)]}_{= H(\pi_\theta(\cdot | s_t))} \right], \end{aligned}$$

because the inner expectation is, by definition, the entropy of the conditional distribution at s_t . Collecting the terms back under a single expectation gives

$$\mathcal{H}(\pi_\theta | x) = \mathbb{E}_{y \sim \pi_\theta} \left[\sum_{t=1}^T H(\pi_\theta(\cdot | s_t)) \right]. \quad (5)$$

Equation (5) is an identity, not an approximation. Its left-hand side is what we want to protect; its right-hand side is what instrumentation can measure, since $H(\pi_\theta(\cdot | s_t))$ is read directly off the logits at each position.

The entire argument of this paper follows from one feature of Eq. (5) that is easy to pass over: *the measure $\mathbb{E}_{y \sim \pi_\theta}$ depends on θ as well*. The right-hand side is not a sum of θ -dependent terms over a fixed set of positions — it is a θ -dependent average of them over a θ -dependent set of trajectories.

4.2 Step 2: the product rule

Write the per-trajectory total

$$f_\theta(y) := \sum_{t=1}^T H(\pi_\theta(\cdot | s_t(y))), \quad (6)$$

so that Eq. (5) reads $\mathcal{H} = \sum_y \pi_\theta(y | x) f_\theta(y)$. Both factors of every summand carry θ , so differentiating requires the product rule:

$$\begin{aligned} \nabla_\theta \mathcal{H} &= \sum_y [\nabla_\theta \pi_\theta(y | x)] f_\theta(y) \\ &\quad + \sum_y \pi_\theta(y | x) \nabla_\theta f_\theta(y). \end{aligned} \quad (7)$$

The second sum is already an expectation. The first is not, because $\nabla_\theta \pi_\theta$ is not a probability distribution; the score-function identity $\nabla_\theta \pi_\theta = \pi_\theta \nabla_\theta \log \pi_\theta$ restores it to one. Substituting and naming the two terms,

$$\begin{aligned} \nabla_\theta \mathcal{H} &= \underbrace{\mathbb{E}[\nabla_\theta f_\theta(y)]}_{\text{(L) local}} \\ &\quad + \underbrace{\mathbb{E}[f_\theta(y) \nabla_\theta \log \pi_\theta(y | x)]}_{\text{(V) visitation}}. \end{aligned} \quad (8)$$

Equations (3)–(8) contain no approximation of any kind. The split into two channels is forced by the product rule, not chosen.

4.3 Reading the two terms

(L), the local channel, holds the visited states fixed and asks how the conditional distribution at each one changes. Equation (1) is a first-order estimate of exactly this.

(V), the visitation channel, holds the per-state entropies fixed and asks how the probability of reaching each state changes. An update that concentrates mass on one continuation makes the subtree below it more likely and the sub-trees below its siblings less likely. Nothing in Eq. (1) can see this: Ω reads the conditional distribution at a position and never reads the probability of arriving there.

The two terms can have opposite signs. A method that drives (L) to zero has not thereby bounded $\nabla_\theta \mathcal{H}$.

4.4 Why the local channel cannot be repaired

A natural objection is that the two channels might coincide in practice: branch points are positions where the policy is uncertain, so they are high-entropy positions, so Ω should be large there and

the local term should protect them automatically. The premise is correct — in warm-up rollouts on our model, branch diversity correlates with local entropy at Spearman $\rho = 0.65$, and with the entropy-to-go at $\rho = 0.82$. The inference from it fails, in three steps.

Step 1: Ω reads a deviation, not a level. Collecting the constants of Eq. (1) into $c := \eta \mathbb{I}_{\text{clip}} A / (L \pi_{\text{old}})$ and rewriting in terms of surprisal gives, for the contribution of a single action a ,

$$\Omega \propto c \pi_a (1 - \pi_a) (H - I_a), \quad (9)$$

and $H = \mathbb{E}_{a \sim \pi} [I_a]$ by definition. The factor $H - I_a$ is therefore a mean-zero fluctuation: what sets the magnitude of $|\Omega|$ is the *spread* of surprisal, not the *height* of the entropy. Nothing in c depends on the shape of $\pi(\cdot | s)$, so the argument below is unaffected by the clipping indicator or the importance weight.

Step 2: at a uniform branch point $\Omega \equiv 0$. If the distribution is uniform over k candidates then $I_a = \log k = H$ for every a , so Ω vanishes identically. The positions that are most branch-like — several continuations at comparable probability — are exactly where the local term contributes nothing.

Step 3: and that zero is the correct answer. This is the step that closes the argument, and it follows from Hao et al.’s own derivation. Differentiating the conditional entropy with respect to a single logit gives (Hao et al., 2026, Eq. 20)

$$\frac{\partial H(\pi_\theta | s)}{\partial z_{s,a}} = -\pi_\theta(a | s) (\log \pi_\theta(a | s) + H(\pi_\theta | s)), \quad (10)$$

and at a uniform distribution $\log \pi_\theta(a | s) = -\log k = -H$ makes the bracket vanish for every a . So $\partial H / \partial z_a = 0$ identically: Ω is not wrong, it is right. Perturbing one logit of a uniform distribution by δ and writing $S := e^\delta + k - 1$ gives the closed form

$$H(\delta) = \log S - \frac{\delta e^\delta}{S}, \quad H''(0) = -\frac{k-1}{k^2}, \quad (11)$$

with $H'(\delta)$ of sign $-\text{sign}(\delta)$: any push in any direction lowers the entropy, but the loss is entirely second order. For a 10-way uniform distribution and $\delta = 0.1$ the actual entropy loss is 4.7×10^{-4} , roughly two orders of magnitude below that of a

distribution	H	$P(\text{split})$	$\mathbb{E} \Omega / A $	$\mathbb{E}[1-\pi_a]$
$q=.99$	0.10	0.08	0.092	0.020
$q=.9, \text{ tail}$	1.25	0.57	1.129	0.190
$q=.6, m=3$	1.11	0.98	0.457	0.587
unif. $k=10$	2.30	1.00	0.000	0.900
unif. $k=100$	4.61	1.00	0.000	0.990

Table 1: The local channel and the visitation channel are ordered oppositely. Distribution family: top token has mass q , the remaining $1 - q$ is spread uniformly over m tokens. $P(\text{split})$ is the probability that $n=8$ samples do not all agree. $\mathbb{E}|\Omega|$ is taken over the token actually sampled at that position.

peaked distribution under the same perturbation. As far as local entropy is concerned, uniform branch points are genuinely safe, and a local estimator that allocates them no protection is allocating correctly for its own objective. No refinement of the local term recovers the missing signal, because the missing signal is not local.

The two channels are anti-correlated. The visitation term’s prefactor, derived in Section 7, is $1 - \pi_a$, which approaches 1 as the distribution flattens — and the probability that a group of n samples actually diverges at a position approaches 1 in the same limit. Table 1 evaluates both channels on a family of distributions that interpolates between confident and flat. $|\Omega|$ is dominated by positions where the model is fairly confident but a tail token was sampled; at those positions siblings diverge only about half the time. Where divergence is certain, the local term is identically zero and the visitation prefactor is maximal.

The statement we can therefore make is stronger than “the local term is incomplete”: *the local term is anti-correlated with the term it omits*, and the omission cannot be fixed within the local channel.

5 Estimating the Visitation Channel

5.1 Causal elimination gives a reward-to-go

The visitation term of Eq. (8) is not yet in a usable form: the factor $f_\theta(y)$ multiplying the score is a property of the *whole* trajectory, so a naive estimator would credit every token with entropy that appeared before it. Causality removes those terms exactly.

Expanding both factors — f_θ by Eq. (6) and the score by $\nabla_\theta \log \pi_\theta(y | x) = \sum_u \nabla_\theta \log \pi_\theta(y_u |$

s_u) — turns (V) into a double sum:

$$(V) = \sum_t \sum_u \mathbb{E}[H(\pi_\theta(\cdot | s_t)) \cdot \nabla_\theta \log \pi_\theta(y_u | s_u)]. \quad (12)$$

Consider a term with $u \geq t$. The entropy $H(\pi_\theta(\cdot | s_t))$ depends on y only through $y_{<t}$, and $y_{<t} \subseteq y_{<u}$, so it is measurable with respect to $y_{<u}$ and may be pulled out of a conditional expectation given $y_{<u}$. What remains is the expected score at a single position, which vanishes for any distribution:

$$\begin{aligned} & \mathbb{E}[\nabla_\theta \log \pi_\theta(y_u | s_u) | y_{<u}] \\ &= \sum_a \pi_\theta(a | s_u) \frac{\nabla_\theta \pi_\theta(a | s_u)}{\pi_\theta(a | s_u)} \\ &= \nabla_\theta \sum_a \pi_\theta(a | s_u) = \nabla_\theta 1 = 0. \end{aligned} \quad (13)$$

By the tower rule every (t, u) term with $u \geq t$ is therefore *exactly* zero — not small, zero. This is the same identity that licenses a baseline in policy gradients, used here to enforce causality rather than to reduce variance.

Only $u < t$ survives. Swapping the order of summation collects, for each token u , the entropies at all later positions:

$$(V) = \mathbb{E}\left[\sum_u \nabla_\theta \log \pi_\theta(y_u | s_u) H_{\text{togo}}(s_u \oplus y_u)\right], \quad (14)$$

with $H_{\text{togo}}(s_u \oplus y_u) := \sum_{t>u} H(\pi_\theta(\cdot | s_t))$.

We stress that H_{togo} was not posited. Equation (14) is a REINFORCE estimator whose reward is “the conditional entropy remaining after this token”, and H_{togo} is that reward-to-go.

5.2 The sibling-prefix baseline

Any function $b(s_u)$ of the state alone leaves Eq. (14) unbiased, by the same argument. Subtracting one gives the per-token advantage

$$A_H(s_u, y_u) := H_{\text{togo}}(s_u \oplus y_u) - \bar{H}_{\text{togo}}(s_u), \quad (15)$$

and up to this point *no approximation has been made*.

The baseline we use is the mean of H_{togo} over the rollouts in the same prompt group whose prefixes $y_{<u}$ agree exactly. Those rollouts share the state s_u by definition, so their mean is a function of s_u alone and the baseline is admissible. The condition is agreement on $y_{<u}$ and not on $y_{\leq u}$: the

rollouts must diverge *at* u for the difference to be a branch score.

Two properties follow immediately and matter later. First, the implementation includes a rollout in its own sibling set, so b depends weakly on y_u and the estimator is not strictly leave-one-out; the resulting bias is $O(1/m)$ in the number m of surviving siblings, the same property GRPO’s group-mean advantage has. Second, when $m = 1$ the baseline equals the rollout’s own value and $A_H \equiv 0$, so a position with no siblings is silent rather than noisy.

6 Four Approximations

Equation (15) is exact but not computable: H_{togo} sums conditional entropies at states the update has not visited. Three approximations make it computable and a fourth bounds its influence.

6.1 A1: horizon truncation and discounting

We truncate the sum to κ steps and discount:

$$H_{\text{togo}}(s_u \oplus y_u) \approx \sum_{k=1}^{\kappa} \gamma_H^k H(\pi(\cdot | s_{u+k})). \quad (16)$$

The discount serves two purposes: distant terms carry more forecast error (A2) and therefore deserve less trust, and the sum stays bounded independently of κ .

What this discards — the tail $\sum_{k>\kappa}$ and a fraction of the retained terms — is largely harmless for a reason specific to our use. Equation (15) consumes only the *difference* between siblings. Any bias common to rollouts sharing s_u cancels in that difference, and truncation bias is common to first order.

6.2 A2: forecasting the unrealized future

Even Eq. (16) requires the future states s_{u+k} . At update time they do not exist: rollouts came from π_{old} , each state has exactly one sampled continuation, and re-rolling is prohibitive. We replace the realized future with a parallel forecast from the current hidden state h_u :

$$H(\pi(\cdot | s_{u+k})) \longrightarrow H(p_{\text{MTP}}^{(k)}(\cdot | h_u)). \quad (17)$$

Head k is a two-layer residual MLP applied to h_u , followed by the policy’s own unembedding; the heads own no vocabulary parameters. Their last layer is zero-initialized, so before any training every head reproduces the policy’s own next-token distribution and the forecast starts from a

sane value. In the RL path the logits are materialized in chunks and reduced to an entropy immediately, so the $[\kappa, B, T, V]$ tensor never exists.

The error of this substitution has a known sign. Conditioning on the intervening tokens is exactly what the forecast omits. Fixing S_u , the k -th term of Eq. (16) has expectation $H(Y_{u+k} | Y_{u+1:u+k-1}, S_u)$, whereas the MTP head models the *marginal* $p(Y_{u+k} | S_u)$. Since conditioning reduces entropy,

$$\begin{aligned} H(Y_{u+k} | S_u) - \mathbb{E}[H(\pi(\cdot | S_{u+k-1}))] \\ = I(Y_{u+k}; Y_{u+1:u+k-1} | S_u) \\ \geq 0. \end{aligned} \quad (18)$$

The forecast *always over-estimates*, and the excess is precisely the mutual information with the intervening tokens, which grows monotonically with k .

Equation (18) is not merely a bound: it is the reason the marginal is the quantity we want. It is the fan-out of the continuation — how many ways the next k tokens can unfold — rather than the residual uncertainty along the one path that happened to be sampled.

6.3 A3: per-head affine calibration

Equation (18) predicts a per-head, monotone inflation, so we absorb it with a per-head affine correction fitted by least squares against oracle entropies on warm-up rollouts:

$$\hat{H}_k = a_k H(p_{\text{MTP}}^{(k)}(\cdot | h_u)) + b_k. \quad (19)$$

The fitted coefficients confirm the prediction. On our model the first head has $a_1 = 1.03$ — a one-step forecast has no intervening tokens, so the mutual information in Eq. (18) is zero and the approximation is exact — while a_k drops to 0.10–0.19 for $k \geq 2$. Calibration switches the unreliable heads off on its own.

Combining A1–A3 gives the forecast that is actually computed:

$$\begin{aligned} \hat{H}_{\text{togo}}(s_u \oplus y_u) = \\ \max\left(0, \sum_{k=1}^{\kappa} \gamma_H^k (a_k H_k(h_u) + b_k)\right), \end{aligned} \quad (20)$$

the clamp encoding the fact that an entropy cannot be negative. With $\kappa = 2$ and $\gamma_H = 0.7$ this instantiates to $\hat{H}_{\text{togo}} = 0.72 H_1 + 0.06 H_2 + 0.11$:

the second head contributes about 8%, so the effective horizon is close to one step. We regard this as a consequence of Eq. (18) rather than a tuning artifact, and note it as a limitation — a forecaster whose error did not grow with k would leave a longer horizon usable.

Alignment. One detail decides whether the signal is a branch score at all. The hidden state at position u is conditioned on s_u and predicts y_u ; it does not know y_u . The quantity we need, $H_{\text{togo}}(s_u \oplus y_u)$, must therefore be read from position $u + 1$ and shifted back by one. Omitting the shift measures the value of the state *before* the branch rather than the value of the branch taken.

7 From Gradient to Token Weights

Equation (15) is a term to add to a gradient, but methods in this family do not add gradients — they re-weight a loss. What is needed is a first-order prediction of how much one update, acting through token u , changes the trajectory entropy.

How far one step moves a logit. We reuse the logit-space update of Hao et al. (2026, Eq. 23) rather than rederiving it. With $w_u = \text{clip}(r_u, 1 - \epsilon_{\text{lo}}, 1 + \epsilon_{\text{hi}})A_u$ the coefficient the policy loss actually applies, Assumption 2 makes a single step move only the sampled token’s logit, $\Delta z_{u,a} = \eta w_u \delta_{a,y_u} + O(\eta^2)$. The softmax Jacobian $\partial \log \pi_a / \partial z_a = 1 - \pi_a$ then gives

$$\widehat{\Delta \log \pi}_u = \eta w_u (1 - \pi_u), \quad (21)$$

which vanishes as $\pi_u \rightarrow 1$: a token that already holds the mass cannot be given more. This is the prefactor whose anti-correlation with $|\Omega|$ Table 1 records.

The visitation-channel prediction. Because $\nabla_{\theta} \log \pi_u \cdot \Delta \theta = \Delta \log \pi_u$, substituting into Eq. (15) gives the per-token quantity

$$\text{visit}_u := \widehat{\Delta \log \pi}_u \cdot \text{clip}(A_H(s_u, y_u), -c, +c). \quad (22)$$

A4 is the clip, and it is applied to A_H *before* the product rather than after, so that one token’s influence is bounded by $|\eta w_u| \cdot c$. Clipping the product instead would let forecast error be amplified without bound at tokens with large w_u .

Sign quadrants. Equation (22) defines the collapse channel precisely. When $\widehat{\Delta \log \pi_u} > 0$ and $A_H > 0$, mass moves toward a branch richer than its siblings and trajectory entropy rises. When $\widehat{\Delta \log \pi_u} > 0$ and $A_H < 0$, mass moves into a dead end and the diversity carried by the abandoned siblings is destroyed — this is the collapse path that (L) is structurally unable to see. Negative $\widehat{\Delta \log \pi_u}$ reverses both readings.

8 Combining the Channels

The visitation term could be added to Ω before the weight map, but measured on real rollouts that map destroys the signal: Ω carries a factor $1/\pi_{\text{old}}$, so $\max |\Omega|/\text{median} |\Omega|$ reaches 3×10^4 and the band is spent on one or two outliers while the bulk lands on a point. We therefore apply the correction *after* the map, as a bounded adjustment of the weights the base method has already produced.

Let $\mathcal{S} = \{u : \text{visit}_u \neq 0\}$ be the support — by Eq. (24), the branch points — and let $b = \lambda_{\max} - \lambda_{\min}$ be the width of the weight band. For $u \in \mathcal{S}$,

$$\alpha_u^f = \text{clamp}\left(\alpha_u + \lambda b \tanh(\text{visit}_u/\rho), \lambda_{\min}, \lambda_{\max}\right), \quad (23)$$

with $\rho = \text{rms}\{\text{visit}_v : v \in \mathcal{S}\}$; $\alpha_u^f = \alpha_u$ elsewhere.

Three properties matter. The $\tanh$ is odd, monotone and bounded, so it preserves the sign of the visitation term, caps one token’s movement at λb , and does not depend on the absolute scale of visit. The RMS is taken over $\mathcal{S}$ alone rather than over the whole response: normalizing by the whole response would shrink the correction in proportion to how sparse branch points are, which is a property of the sampler and not of the signal. And at $\lambda = 0$ the correction vanishes identically, so the pipeline is bitwise identical to the base method — a regression test enforces this.

Because the map runs first and the correction second, the step size η of Eq. (21) enters only through visit_u/ρ , where it cancels. We fix $\eta = 1$ and sweep λ alone.

The support is part of the treatment. $\mathcal{S}$ is defined by $\text{visit}_u \neq 0$ rather than by a threshold, deliberately: it selects both where the correction lands and, through ρ , how large every correction is, so a tighter test would change what is trained rather than what is logged. One consequence is

that $\mathcal{S}$ admits float32 round-off — A_H is a deviation from a mean over token-identical siblings, and $\sum_j g_j/m$ does not reproduce g_i bitwise, leaving $|A_H| \sim 6 \times 10^{-8}$ where it is interpretively zero. Those residuals are $\sim 10^{-6}$ relative to a real branch score and cannot move a weight, but they are counted in the support statistics of Section 12.6, which should be read as an upper bound on the branch-point fraction.

9 Tree-Structured Rollouts

9.1 Why plain rollouts cannot support A_H

$\hat{H}_{\text{togo}}$ is a deterministic function of the causal prefix, so two rollouts sharing $y_{\leq u}$ produce identical forecasts. Combined with Eq. (15) this gives a sharp characterization:

$$A_H[i, u] \neq 0 \iff \text{siblings share } y_{<u} \text{ but diverge at } y_u, \quad (24)$$

that is, u is a branch point of the group. Under independent sampling from a sharpened policy, group members diverge at their first token and never share a prefix again, so $m = 1$ almost everywhere and $A_H \equiv 0$: the visitation channel is not merely weak, it is identically absent.

We therefore schedule the sharing. A tree sampler expands a single root and cuts it at fixed depths $d_1 < d_2 < d_3$ with branching factor 2, yielding $n = 8$ leaves whose sibling structure is known in advance. The cuts create the only positions at which Eq. (24) can fire.

9.2 A ceiling on the support

It is tempting to treat the fraction of positions with $A_H \neq 0$ as a quantity to be maximized. It is not, and the reason is combinatorial rather than empirical. Two quantities behave very differently under the tree:

- the fraction of positions where *some* other rollout shares $y_{<u}$ is moved by the cuts, and equals d_L/T by construction;
- the fraction where $A_H \neq 0$ is *not*, because by Eq. (24) it counts branch points, and n sequences form a trie with at most $n - 1$ internal branching nodes.

Since the one-position alignment shift widens each branch node into two columns, the fraction of positions carrying a non-zero A_H is bounded by $2(n - 1)/T$ for *any* sampler whatsoever. With

$n = 8$ and responses of $\sim 10^3$ tokens this ceiling is on the order of a percent. The tree’s role is to make the branch points exist and to place them at controlled depths, not to make them numerous.

9.3 The correction cannot move the mean

A_H is a deviation from a sibling mean, so on each sibling set

$$\sum_{j \in \text{sib}(s_u)} A_H(s_u, y_u^{(j)}) = 0. \quad (25)$$

The correction in Eq. (23) therefore cannot change the average weight assigned to branch tokens. It can only spread siblings apart. This is not a defect to be engineered away: an unbiased estimator requires a baseline, and a baseline forces the mean to zero.

The consequence for evaluation is direct, and we state it before presenting any experiment. “Mean entropy at branch points increased” is *not* a prediction of this method and cannot serve as evidence that it works; the corresponding prediction is that the *variance across siblings* widens. A method designed to raise the mean is a different and weaker hypothesis — it does not need a forecast at all, only a support — and we include it below as a control arm.

10 The Approximation Ledger

Table 2 collects every departure from Eq. (8).

The strongest assumptions are A2 and B2. A2 has a known sign and a known monotonicity (Eq. 18), which is what makes the affine correction in A3 an appropriate remedy rather than a curve fit. B2 — that updating one token does not change the conditional distribution at other positions — is false for a network with shared parameters and has no corrective mechanism. It is not a risk this work introduces, since the base method already relies on it, but adding a second channel adds a second copy of its error.

11 Experimental Setup

11.1 Model, data, and optimization

Within a backbone, all runs fine-tune the same base model on the same data with the same optimizer settings, and differ only in the intervention. Our primary backbone is a 1.5B-parameter mathematics-specialized model; the full five-arm decomposition and every ablation are run there. Three further backbones (Table 3) carry the headline three-way

contrast only, which is what a second model can answer: not *how* the effect decomposes—that is a question about the method, and it is settled once—but whether its *sign* survives a change of scale and of pre-training family. All runs train on the same 17k-problem competition-mathematics set. With a training batch of 512 prompts, 110 optimization steps correspond to about 3.2 epochs.

11.2 Validation set, and why it is not the same everywhere

The mathematics-specialized backbones validate on AIME 2024 (30 problems, 32 samples each). AIME 2024 is small: with a per-problem accuracy spread of .1542 measured across our runs, the standard error of its mean is $.1542/\sqrt{30} \approx .0281$. For a backbone that scores near zero on it, every arm lies inside one standard error, so the benchmark cannot separate them.

The sharper problem is not the reported number but the *selection*: we choose each run’s checkpoint by its best validation accuracy, so a benchmark that cannot distinguish arms also cannot distinguish checkpoints, and the selection becomes a draw from noise. The non-mathematics backbones therefore validate and select on MATH500 (500 problems, one sample each), whose standard error is roughly four times smaller.

This buys a measurable comparison at the cost of a protocol that is not identical across backbones. We state it here rather than in a footnote because it constrains what may be read off the results: every contrast we report is *within* a backbone, and no claim compares absolute accuracy between rows of Table 3.

Base or instruction-tuned. The same measurement forces a second protocol difference. The Llama backbone is the instruction-tuned checkpoint; the other three are base models. We began with the base one and scored it under this protocol before training it: it reached 0.020 on MATH500 and 0.024 on grade-school arithmetic. The cause is mechanical rather than mathematical—that checkpoint ships a chat template, which our data pipeline applies, so a model that was never instruction-tuned receives the instruction format and continues the prompt instead of answering it.

The instruction-tuned checkpoint of the same family and the same parameter count, scored under the identical protocol, reaches 0.340 and 0.803 on those two benchmarks. Holding family and scale

	approximation	what it discards	sign / bound	mitigation
—	Eqs. (5)–(15)	nothing — identities	—	—
A1	horizon κ , discount γ_H	the tail $\sum_{k>\kappa}$, and a fraction of the retained terms	under-estimate	the common component cancels in the sibling difference
A2	realized future $\rightarrow$ MTP marginal	mutual information with intervening tokens, Eq. (18)	over-estimate, monotone in k	A3
A3	per-head affine calibration	non-affine residual of A2	least-squares residual	oracle-forecast control arm
A4	clip A_H to $\pm c$	magnitude of extreme branch scores	bounded by c	saturation rate is logged
B1	first order in η	$O(\eta^2)$	—	small learning rate
B2	only the sampled logit moves	cross-position effects of weight sharing	—	<i>inherited from the base method</i>
B3	rollouts come from π_{old}	mismatch with the π_θ expectation	—	importance ratio restores r
B4	self included in sibling set	$O(1/m)$ relative to leave-one-out	—	same property as the GRPO advantage

Table 2: Every approximation between the exact decomposition and the implemented weight. A1–A4 are introduced by this work; B1–B4 are shared with the token-level entropy-control method we extend.

Backbone	Family	Math-spec.	Validation
1.5B	Qwen2.5-Math	yes	AIME 2024
7B	Qwen2.5-Math	yes	AIME 2024
3B*	Llama 3.2	no	MATH500
7B	Mistral	no	MATH500

Table 3: Backbones. The validation set differs by backbone, deliberately; Section 11.2 says why, and the analysis never compares absolute accuracy across rows. *This row uses the instruction-tuned checkpoint; the other three are base models, and Section 11.2 says why this one cannot be.

fixed and changing only the checkpoint moves MATH500 by a factor of seventeen, which is what identifies the failure as one of format rather than of capacity: this backbone can do the arithmetic, and the base checkpoint was not being asked for it in a form it could answer.

What this costs is not a weak row but an absent one. A group of $n = 8$ responses receives zero advantage when its members score alike, so at that solve rate $(1 - 0.020)^8 \approx 0.85$ of groups carry no gradient at all, on a training set harder than the benchmark just quoted. Neither our method nor the baselines can learn from the remainder, and the row would compare two untrained models. We therefore gate a backbone on the fraction of non-degenerate groups on the *training* set, measured against a backbone known to train, rather than on a validation score: a validation set says whether a backbone can be measured, not whether it can learn. On that gate the instruction-tuned checkpoint leaves — of groups informative against — for the 1.5B mathematics backbone, a ratio of —.

Optimization uses GRPO with $n = 8$ responses per prompt, a constant learning rate of $1e-6$ with no warm-up (so that the schedule is independent of the declared total step count), a maximum response length of 3072 tokens, no entropy bonus, and no KL loss. The clipped surrogate is asymmetric, $\varepsilon_{\text{low}} = 0.2$ and $\varepsilon_{\text{high}} = 0.28$, with the dual-clip lower bound at $c = 10$; we state this because “GRPO” with a symmetric $\varepsilon = 0.2$ and a KL term is a different baseline, and because the follow-up arm of Section 12.10 is defined by moving these two numbers. Every arm, the baseline included, uses the same three values: the only configuration difference between the GRPO arm and STEER-F is the loss mode and the method hyperparameters listed next. Rollouts are sampled at temperature 1.0 with no nucleus truncation; validation uses temperature 1.0 and top- $p = 0.7$. Every arm uses the same data seed and the same initial checkpoint, which we verify by requiring the step-0 validation scores to agree.

11.3 Method hyperparameters

The forecaster uses $\kappa = 2$ heads with discount $\gamma_H = 0.7$ and the per-head calibration of Eq. (19). The combination uses $\lambda = 0.25$, RMS normalization, clip $c = 1.0$, and the sibling-prefix baseline. The weight map is the base method’s exponential map of Eq. (2), taken un-rescaled, at $\lambda_{\text{min}} = 0.7$, so weights land in $[0.7, 1]$ and can only attenuate. The released configuration of the base method sets $\lambda_{\text{min}} = 0.8$; every other value in that launcher matches it. We state plainly how ours was chosen, because it bears on which contrasts can be

arm	λ	rollout	signal in the weight
GRPO	—	plain	none
STEER	0	plain	local Ω only
uniform	.25	tree	support only
permuted	.25	tree	A_H shuffled
STEER-F	.25	tree	A_H as derived

Table 4: The five arms. UNIFORM and PERMUTED run the full pipeline with the forecast’s value destroyed in two different ways, so any gain they show is attributable to the apparatus rather than to the visitation signal.

read: 0.7 was selected on *our* method, not on the base method, so the baseline runs at a damping strength picked for the treatment and 1.5 times the maximum attenuation its own release uses. Section 12.10 reports the base method at $\lambda_{\min} = 0.8$ for this reason.¹ The tree sampler uses a single root, cuts at depths {64, 192, 384}, and branching factor 2 at each cut, giving $n = 8$ leaves.

11.4 Five arms

The design isolates one question: does the *value* of the forecast matter, or only the apparatus that carries it? A comparison between the base method and the full method changes two things at once (λ and the sampler), so it cannot answer this. We therefore add two arms that keep the entire apparatus — the tree, the forecaster, the support, the weight mapping — and destroy only the information in A_H .

UNIFORM applies the same attenuation to every position in the support, discarding A_H ’s value but keeping its support — this is the implementation of the “raise the mean” hypothesis that Section 9.3 distinguishes from ours. PERMUTED keeps the multiset of A_H values within each sibling set and shuffles the assignment, so the support, the magnitudes, and Eq. (25) all survive while the pairing between a branch and its score is destroyed. Together they bracket the claim: if the visitation signal is doing the work, both controls should behave like the no-forecast arms.

11.5 Metrics

We report acc@32 (mean accuracy over 32 samples) and maj@32 (accuracy of the majority vote over the same 32 samples), and their difference

$$\text{uplift} := \text{maj@32} - \text{acc@32},$$

¹Our implementation exposes an optional reshaping of $\tilde{\Omega}$ before this map (winsorizing or rank-transforming it); every run reported here uses the identity setting, so the base method’s map sees exactly what it would have seen.

which measures how much better a set of 32 samples is than its average member — that is, whether correct answers concentrate while incorrect ones scatter. We use uplift rather than pass@32 because the latter does not separate the arms.

Because single validation points are noisy on a 30-problem set, the primary statistic is the mean over the converged window, steps 40–110 (8 validation points per arm). We quote differences in units of the across-problem standard error, $\text{SE} = \text{std@32}/\sqrt{30} \approx 0.027$, which is the correct scale for generalization to new problems. We separately note that the pure re-sampling noise of the protocol — estimated from two validations of identical weights — is about 0.004 in accuracy, an order of magnitude smaller; the two scales answer different questions and we are explicit about which is in use.

One instrumentation caveat propagates into every entropy number we report. The logged `actor/entropy` is aggregated with a token-mean, so it is $\mathcal{H}$ divided by the mean response length, not $\mathcal{H}$. Since response lengths differ by under 5% across arms, we report the raw quantity and, where the distinction could matter, the length-corrected product as well.

12 Results

All numbers below come from the runs described in Section 11: one benchmark, 110 optimization steps per arm, and the same initial checkpoint, whose step-0 accuracy is identical across arms. Every figure is recomputed from a training log released with this paper. We report our own measurements only; no figure in this section is taken from prior work.

The campaign is not finished. Each arm below is averaged over the seeds *it* has finished, a count that differs by arm and is printed as n in every table, while each contrast is computed on the seeds its own two arms share. The two are therefore not interchangeable: a difference of two rows of Table 6 is not the corresponding entry of Table 8, and is not meant to be. Section 12.8 reports what the repeats now measure — the run-to-run spread, and the machine each seed landed on.

Which result carries which weight. What follows is not one kind of result, and the order is the order of how much each one bears. Section 12.3 comes first. It tests a prediction the construction makes before any run — that a correction this sparse and this mean-preserving cannot move an

step	GRPO	STEER	UNIF	PERM	STEER-F
0	.039/.035	.043/.040	.039/.035	.039/.035	.039/.035
10	.075/.127	.064/.109	.070/.124	.062/.126	.078/.148
20	.120/.176	.114/.195	.102/.171	.087/.155	.103/.181
30	.108/.195	.135/.215	.121/.223	.099/.178	.113/.211
40	.144/.236	.115/.187	.119/.201	.116/.197	.141/.230
50	.120/.204	.125/.196	.121/.216	.153/.224	.145/.258
60	.126/.178	.139/.202	.125/.203	.135/.190	.145/.237
70	.130/.200	.128/.200	.129/.197	.132/.227	.147/.257
80	.135/.182	.135/.202	.140/.204	.134/.197	.144/.206
90	.136/.191	.144/.212	.142/.178	.130/.210	.166/.239
100	.132/.171	.146/.207	.134/.192	.144/.212	.157/.222
110	.157/.208	.132/.181	.140/.171	.152/.218	.151/.230

Table 5: AIME24 acc@32/maj@32 at every validation point. Rows below the rule are the converged window used for all subsequent statistics.

average taken over every response token — and it tests it as a contrast *between groups of contrasts*, on up to 4 seeds, with the two groups’ $|t|$ not overlapping. That is the structural claim, and it is what this campaign establishes. The accuracy contrasts after it rest on 3 paired seeds each, and they do not all survive: the gap against STEER clears a conventional threshold, the gap against PERMUTED is marginal, and the headline gap against GRPO is smaller than GRPO’s own spread across its seeds (Section 12.8) and does not. One pre-registered falsification condition is met at this seed count, on the contrast that isolates A_H ’s value; Section 12.4 states which and what it leaves standing. A reader who wants one result from this section should take the dissociation, not the gap.

12.1 Validation trajectory

Table 5 gives AIME24 accuracy and majority-vote accuracy at every validation point. The four arms that share the tree pipeline start from an identical step-0 score; STEER starts at .043 rather than .039, a gap we attribute to re-sampling noise rather than to a different checkpoint, since two validations of the *same* base weights in our logs differ by .004 in accuracy.

12.2 Converged window

Table 6 averages each arm over steps 40–110.

Two orderings are worth stating before anything is subtracted. On accuracy all three arms that do not carry A_H ’s value — STEER, UNIFORM, PERMUTED — sit *below* plain GRPO, and STEER-F is the only one above it. It is above by less than GRPO’s own standard error across seeds, which is why the ordering and not the size is what we take

arm	n	acc@32	maj@32	uplift	entropy
GRPO	3	.1436 ± .0064	.2145 ± .0100	.0710	.1575
STEER	3	.1371 ± .0021	.2090 ± .0054	.0719	.1155
UNIFORM	3	.1422 ± .0057	.2114 ± .0093	.0692	.1340
PERMUTED	3	.1405 ± .0030	.2129 ± .0023	.0724	.1219
STEER-F	3	.1491 ± .0005	.2301 ± .0042	.0810	.1263

Table 6: Means over the converged window (steps 40–110, eight validation points per run), every arm averaged over the same $n = 3$ seeds — the ones all five arms have finished — with the standard error across them. Because the seed set is shared, a difference of two rows here *is* the paired contrast of Table 8 for every contrast drawn on those seeds: STEER-F – GRPO reads +.0055 either way. The two exceptions are the contrasts whose arms both reach a seed this table drops, which pair on 4 seeds instead and therefore differ a little from the row difference; Table 8 carries n per row. Restricting to the shared seeds also equalizes the machine: each arm here is one two-GPU seed and two four-GPU seeds, so a row difference no longer carries a box mix (Section 12.8). The $\pm$ is a standard error over 3 draws, not a precision — STEER-F’s three land within .0016 of one another, which is agreement worth noting and still three draws; read it against the run-to-run ranges in Section 12.8. Every seed, including the ones outside the shared set and the one held out for not finishing, is in the released `per_seed.tsv`. *uplift* is `maj@32 – acc@32`. Entropy is the token-mean quantity discussed in Section 11, i.e. $\mathcal{H}$ divided by mean length.

from this table. On entropy GRPO holds the most by a clear margin and the four intervened arms sit close to one another, STEER-F among them, so the two columns are neither aligned nor reversed. Neither ordering is the instrument for either question at this seed count, and the next section says why: read as contrasts rather than as a ranking, the entropy column tracks the damping apparatus and the accuracy column tracks A_H .

12.3 Entropy: the aggregate moves with the apparatus, not with A_H

What the method predicts here, fixed in advance.

This method does not set out to hold the aggregate entropy up, and two facts already established above say it cannot. Section 9 bounds the positions at which A_H can be non-zero at $2(n - 1)/T$, about one in eighty; Section 9.3 shows A_H sums to zero over a sibling set, so at the positions it does reach it redistributes weight among siblings rather than adding any. An intervention that is this sparse and mean-preserving by construction should leave an average over every response token close to where it found it. Both statements were written into the manuscript before the campaign was launched,

contrast	n	$\Delta\text{ent.}$	t	Δacc	t
<i>the apparatus changes, A_H does not</i>					
STEER – GRPO	4	-.0491	-5.58	-.0077	-1.98
UNIF – STEER	4	+.0184	+18.52	+.0054	+2.05
<i>A_H changes, the apparatus does not</i>					
STEER-F – UNIF	3	-.0077	-1.42	+.0069	+1.16
STEER-F – PERM	3	+.0044	+2.15	+.0087	+3.11
<i>both change</i>					
STEER-F – STEER	3	+.0108	+1.76	+.0120	+5.24

Table 7: Converged aggregate entropy and accuracy, paired within a seed and grouped by what the contrast changes. Turning the damping apparatus on or off moves the aggregate entropy at $t = -5.58$ and $+18.52$; changing what A_H contains, with the apparatus held fixed, does not move it detectably ($t = -1.42$). Accuracy runs the other way. A row with $n = 1$ has no t .

Section 9.3 explicitly so (“we state it before presenting any experiment”), and the artifact history dates them an hour ahead of the pre-registration discussed below.

What the seeds show. Table 7 groups the contrasts by what each one changes, and the grouping separates them cleanly. The two contrasts that change the apparatus move the aggregate: turning the damping on costs $-.0491$ at $t = -5.58$, and adding the tree with damping applied uniformly returns $+.0184$ at $t = +18.52$ — the latter reproducing across all 4 seeds to within a thousandth. The contrasts that hold the apparatus fixed and change only what A_H contains move it by $-.0077$ ($t = -1.42$) when its value is discarded and $+.0044$ when its pairing is destroyed. Accuracy runs the other way: the apparatus contrasts give $+.0054$ and $-.0077$, while the two A_H contrasts give $+.0069$ and $+.0087$. The quantity that moves the aggregate entropy is the damping apparatus; the quantity that moves accuracy is A_H . That is the predicted pattern, and it is the sense in which this paper is not a method for suppressing entropy.

A pre-registered condition we retire rather than score. The pre-registration (Section C) also fixed a condition on this section: if aggregate entropy orders the arms the same way accuracy does, a dissociation argument fails. We retire that condition instead of reporting it, and say plainly why, since retiring a registered test is exactly the move a reader should be suspicious of. First, it tests the aggregate as a signal about A_H , which the two paragraphs above — and Section 9.3, committed an

hour before the registration was — say it cannot be. Second, an ordering over five arm means is not stable at this seed count: the rank correlation between converged entropy and converged accuracy comes out with a different sign depending on whether each arm is averaged over the seed common to all five (Table 6) or over its own, which differ in number while the campaign runs. Both seed sets are in the released `per_seed.tsv`. Third, and because of the first two, *we claim nothing from it in either direction*: the condition is not reported as passed, and the theory is not reported as damaged by it. It was the wrong instrument to have registered, and the honest correction is to say so and to register a better one.

The measurement that is actually owed. Section 9.3 states the prediction that a sum-zero correction does make: not that entropy at branch points rises, but that the *spread across siblings* widens. No quantity we currently log tests it. The trainer does log an entropy split (`steerf/branch_entropy`), and an earlier draft of this section read it as entropy at branch points; it is not. That monitor takes a fixed top decile of A_H — a tenth of all response tokens — while the positions A_H can reach are the one-in-eighty that Section 12.6 measures, an eighth as many. Since A_H is exactly zero at almost every position, the remainder of that decile is filled by an arbitrary tie-break rather than by branching. Section 12.6 reports what that monitor does and does not select. The replacement restricts the split to $A_H \neq 0$, the same predicate the correction itself uses, and is instrumented in the release; it reports — for the contrast that isolates the pairing, and the cells stay empty until runs carry it. That test is registered in Section C, before those runs, and we state here that it is registered after this section’s original condition was retired rather than before.

What the numbers above cannot carry. The claim is about magnitudes, not about one row failing a threshold. Changing the apparatus moves the aggregate by $+.0184$ and $-.0491$; changing what A_H contains moves it by $+.0044$ and $-.0077$, and the smaller of the two apparatus effects is still more than twice the larger of the two A_H effects. The $|t|$ bands separate the same way and do not overlap — -5.58 and $+18.52$ against -1.42 and $+2.15$ — which is why the argument is the contrast *between groups* rather than any single row. Two cautions travel with it. A small difference that reproduces

contrast	n	Δ_{acc}	t	Δ_{maj}	t
STEER-F – GRPO	3	+0.0055	+0.81	+0.0155	+1.34
STEER-F – STEER	3	+0.0120	+5.24	+0.0211	+2.37
STEER-F – UNIF	3	+0.0069	+1.16	+0.0187	+1.77
STEER-F – PERM	3	+0.0087	+3.11	+0.0172	+2.67
STEER – GRPO	4	−0.0077	−1.98	−0.0052	−1.14
UNIF – STEER	4	+0.0054	+2.05	+0.0014	+0.31

Table 8: Contrasts paired by seed: each arm is averaged over its converged window, arms are paired within a seed, and t is the paired t on the n seed-level differences ($\text{df} = n - 1$). Pairing within a seed cancels any per-seed effect common to all five arms. A contrast’s n is the smaller of its two arms’ seed counts.

across seeds can still reach a large t : the A_H contrast on PERMUTED does, at +.0044, and that is a statement about consistency, not about size. And 3 seeds entitle us to say the aggregate does not move *much* when A_H changes, not that it is unchanged.

12.4 Does the forecast’s value matter?

Table 8 averages each arm over its converged window, pairs the arms within a seed, and reports the mean difference together with a paired t on those seed-level differences. The step-by-step version of this statistic, which the single-seed draft reported, is in Appendix D; it answers a question about one trajectory rather than about re-running.

These contrasts carry less than the dissociation above, and we read them for direction rather than magnitude. They pair on 3 to 3 shared seeds against that section’s 4, and the quantity they measure — an accuracy gap — is the one the seeds in hand do not separate from run-to-run spread (Section 12.8).

The block structure is what the design was built to expose. Every contrast that changes the *value* of A_H is positive; the one that leaves it alone is not distinguishable from zero. UNIFORM runs the entire apparatus — tree sampler, forecaster, support, weight map — and differs from STEER by +.0054 ($t = +2.05$ over 4 seeds): turning the machinery on and feeding it a constant buys nothing. PERMUTED keeps the machinery *and* the magnitudes of A_H and destroys only the pairing between a branch and its score, and STEER-F still exceeds it by +.0087.

Every contrast here now rests on 3 paired seeds, and they do not all say the same thing. Against STEER the gap is +.0120 at $p = 0.035$; against PERMUTED it is +.0087 at $p = 0.090$. Against UNIFORM and against GRPO it is 0.366 and 0.503 — the headline contrast is the weakest row in the table, for the reason Section 12.8 gives: it is the

one that changes three things at once, and GRPO’s own spread across seeds is wider than the effect. The signs agree in the mean everywhere A_H carries information, but the per-seed signs do not, and the next paragraph is about the one place they part.

A pre-registered falsification condition is met.

The pre-registration (Appendix C) named four things that would count against this paper, before any of these runs existed. One of them has happened, and it is the one that clause calls the claim of the paper: *if STEER-F – UNIFORM straddles zero across seeds, the forecast’s value does not matter and the apparatus explains the effect*. On accuracy it straddles. Across the 3 paired seeds the difference runs from −.0020 to +.0182, with 1 of them negative. We report that as a refutation of the registered claim rather than as noise, and we do not rescue it with the mean or with the majority-vote column, where the same contrast happens not to straddle (0 of 3 negative) — the registered primary endpoint is converged-window accuracy.

The other three conditions did not trigger, and the contrast between the first and the second is the informative part. STEER-F – PERMUTED, which holds the magnitudes of A_H and destroys only the pairing, does *not* straddle: +.0033 to +.0125, 0 negative. Aggregate entropy does not order the arms the way accuracy does — STEER-F is third from the top on one column and first on the other in Table 6 — so the dissociation argument of Section 12.3 stands. And the mean token weight stays at .9990, so A_H is sum-zero over sibling sets as derived.

Read together: the *pairing* of A_H to a branch survives its registered test and its *value* does not. The two arms were built to separate exactly those two things, so this is the distinction the design can draw, not an evasion of the result. It leaves the paper with a tension we do not resolve here. The pre-registration calls the UNIFORM contrast the central claim; the argument this paper now leads with is the dissociation of Section 12.3, which on these same 3 seed pairs separates entropy from accuracy at $p = 0.005$ against $p = 0.503$. Both are on the record; a reader who weights the pre-registration above the reframing should read the accuracy claim as refuted at this seed count.

This is the contrast the design was built for. A comparison of STEER-F against STEER alone changes two things at once, and cannot separate the forecast from the sampler that delivers it; the

two destroyed-forecast arms do separate them.

12.5 Where the gain comes from

The majority-vote gain decomposes into a part explained by per-sample accuracy and a part explained by the sample set’s structure:

$$\underbrace{+.0155}_{\Delta_{\text{maj}}} = \underbrace{+.0055}_{\Delta_{\text{acc}}} + \underbrace{+.0100}_{\Delta_{\text{uplift}}},$$

against GRPO; against STEER the uplift term is $+.0091$ ($t = +1.24$ over 3 seeds). Against GRPO roughly two thirds of the improvement is not that individual samples are more often right, but that the right ones agree with each other more and the wrong ones agree less. The two backbones of that split are measured on one seed so far, and the split itself is the claim to re-check as the remaining seeds arrive: against STEER the same term is smaller and well inside its own spread.

Where that structural change comes from is a separate question, and the arm that answers it finished after this section was first written. UNIFORM runs the sampler at $\lambda = .25$ with the forecast’s value destroyed and shows no uplift gain, which reads as though the structure belonged to A_H . It does not: the $\lambda = 0$ tree arm — the sampler with the future channel off, STEER’s local damping still in place — reaches an uplift of .0870, against .0719 for STEER and .0810 for STEER-F. On this evidence most of the structural gain is the sampler’s, and Section 12.10 reports it that way. The decomposition above is unaffected — it is measured against GRPO, which has no tree — but the attribution is not what the five-arm design alone could see.

12.6 Mechanism diagnostics

Section 9 bounds the fraction of positions that can carry a non-zero A_H at $2(n-1)/T$ for any sampler. With $n = 8$ and a mean response length of 984 tokens the bound is .0142. Measured over training, the three tree arms sit at .0136 (UNIFORM), .0129 (PERMUTED) and .0120 (STEER-F), i.e. at 84–96% of a ceiling that no sampler can exceed. A plain-rollout run with the same λ reaches .003. As Section 8 notes, this count admits float32 round-off and is an upper bound; the runs reported here predate the round-off-free counter, so we quote the loose figure and label it as such.

Two readings follow. The tree sampler is close to saturating what the combinatorics allow, so there is little headroom in scheduling more cuts; and

the intervention is extremely sparse — about one position in eighty — which makes the size of the effect in Table 8 the surprising part rather than the small support.

That sparsity also disqualifies one of our own monitors, which we record here rather than leave for a reader to find. The logged entropy split (`steerf/branch_entropy` and its gap) selects the top decile of A_H by rank, a tenth of all response tokens, against a support an eighth that size. A rank-based cut was chosen deliberately — A_H is exactly zero wherever a rollout has become its own only sibling, so a value-based threshold selects almost everything — but with a support this small the decile is filled out by the tie-break among those zeros, and the tie-break is not a random sample of them. A top-decile cut does recall about half of the true divergence points, five times what selecting at random would give, so the bucket is genuinely enriched; it is nonetheless mostly not branch positions, and the mean entropy it reports cannot be attributed to them. Section 12.3 states what replaces it.

12.7 Is the future term a local term?

Section 6 reports a calibration that suppresses every head past the first, leaving $\hat{H}_{\text{togo}}$ dominated by one step ahead. That invites the sharpest objection to the paper: if the forecast reaches one step, is A_H anything a purely local quantity could not have produced? The question is answerable without training anything. We apply the *same* operator — the same sibling baseline, mask and grouping — to two inputs: the calibrated forecast, and the policy’s own next-token entropy read at the same offset, the state after the branch token. Any difference between the two is the forecaster’s contribution and nothing else. We report it on the sibling support, where A_H can be non-zero at all; outside it both are identically zero and a statistic over all positions measures the support rather than the forecast.

The answer is that the forecaster is not earning its cost. Over 464 branch points from 4 validation prompts, the two advantages correlate at $r = +0.917$ (Spearman $+0.937$), the local twin explains $R^2 = 0.840$ of the treatment, and the two agree in sign 0.921 of the time — and the sign is what the token weights are monotone in. The levels correlate less than the advantages ($+0.881$), which is the wrong way round for an artefact of scale.

The mechanism is one the paper had already measured without connecting it. With $\kappa = 2$ and

$\gamma_H = 0.7$ the calibration instantiates to $\hat{H}_{\text{togo}} = 0.72 H_1 + 0.06 H_2 + 0.11$, the sibling baseline removes the constant, and the first head is built to reproduce the policy’s own next-token distribution — its final layer is zero-initialised and its fitted $a_1 = 1.03$ says it stayed there. So A_H should be close to $\gamma_H a_1 = 0.72$ times the local twin, and the measured magnitudes are 0.3445 against 0.4647, a ratio of 0.741. A_H is very nearly a scalar multiple of a quantity the trainer already computes, and Eq. (23) divides the scale out through $\rho = \text{rms}\{\text{visit}\}$.

Two things this does *not* say. It does not touch Eq. (8), which is an identity, nor Section 4.4: the local twin is not Ω . Ω carries no sibling baseline and no tree, vanishes at a uniform branch point, and the twin exists only because the tree supplies siblings to difference against. What the measurement indicts is the *forecasting apparatus* — the MTP heads, their warm-up and their calibration — not the visitation channel. Read alongside Section 12.10, where the sampler alone recovers .1439 against STEER’s .1371 while the forecast adds the remainder, two independent measurements now point the same way.

The caveats are the ones a single diagnostic run carries: 4 prompts at one validation step of the base policy, on independently sampled rollouts, whose sibling support therefore sits early in the response rather than at the scheduled branch depths. The head-free arm of Table 10 is the direct test, and it has now been run. ORACLE reads H_{togo} from the policy’s realised entropy, which verl computes in the same pass, so it carries none of the MTP overhead of Section 12.13 — 416 s per optimization step against STEER-F’s 516 at the same seed on the four-GPU box, 19.3% less. It is also behind it on accuracy. Section 12.10 reports both numbers and what one seed of each can carry.

12.8 Seeds, not steps

Every statistic above treats the seed as the unit of analysis: each run is first collapsed to one number by averaging its converged window, the arms are then paired within a seed, and the test is a one-sample t on the paired differences with $\text{df} = n - 1$. Pairing within a seed matters because a seed carries effects common to all five arms — the data order, and on a split campaign the machine a run landed on — and those cancel exactly in a difference. A contrast’s n is the smaller of its two arms’ seed counts, which is why the table reports n per row

rather than once.

The alternative statistic, a paired t over the eight window steps *within* one run, measures whether a gap is stable along a single trajectory. That is a different question from whether it survives re-running, and we report it only in Appendix D so the two are not confused.

One run is excluded, and the pre-registration says which. STEER-F at seed 4 stopped at training step 91 when its machine lost a GPU, so its window covers six validation points where every other run covers eight. A mean over six points is not a shorter version of a mean over eight; paired against the others it is a different statistic. The pre-registration (Appendix C) excludes a run for mechanical failure and relaunches it with the same seed, so that is what happens here, and the release reports the run with an `partial` flag rather than deleting it. It is counted in the spread below, for the reason given there.

A seed index also names a machine. Seeds 1 and 2 ran on a two-GPU box and seeds 3 and 4 on a four-GPU box, uniformly across arms. The effect is measurable: on *GRPO*, which has finished 2 seeds on the smaller box and 2 on the larger, the two-GPU seeds average .1391 against .1479 on four — a difference of +.0087, larger than the gap between the top two rows of Table 6. Two things keep it out of the comparisons. The contrasts pair inside a seed and therefore inside a machine, so it cancels in the pair. And Table 6 restricts every arm to the seeds all five share, which is one two-GPU seed and two four-GPU seeds for each of them, so the rows carry the same mix rather than different ones. The release prints the per-arm machine mix on every run of the analysis.

Run-to-run spread, and it is the number that governs this section. Every arm has now finished more than once. GRPO’s converged-window accuracy moves from .1350 to .1560 across its 4 seeds, a spread of .0210; STEER-F moves from .1481 to .1497 across its 3 seeds, a spread of .0016; PERMUTED moves only .0095 across 4. Both extremes are measured on the same benchmark with the same protocol, and they differ by more than an order of magnitude.

Set the headline gap of +.0055 against those. **It is a fraction of GRPO’s own spread across seeds and several times STEER-F’s**, so the two arms it compares do not even agree on how large a seed-to-

seed difference is, and each is observed a handful of times. That is the regime in which a single paired difference cannot decide the question, and we do not claim it does. A range between three draws is also not a standard error: the largest of these ranges is more than ten times the smallest, so they do not bound each other and no contrast here should be read against a noise floor quoted to one significant figure. For scale at the other end, two validations of the *same* weights on these 30 questions differ by about .004, so seed variation is the larger term by roughly an order of magnitude and any comparison quoted at single-run precision is quoting noise.

We therefore state what this experiment establishes and what it does not. The mechanism contrasts — what changes when A_H 's value is discarded or its pairing destroyed, with the apparatus held fixed — are the claim. The effect size is what the remaining seeds are for, and on the seeds in hand the headline gap is not separated from the spread.

12.9 Six benchmarks

A single 30-question benchmark cannot separate a method from a lucky draw, so each run's selected checkpoint is re-evaluated on six mathematics benchmarks (Table 9) under the protocol of Section 11: AIME24, AIME25 and AMC23 at avg@32, MATH500, Minerva and OlympiadBench at avg@1.

One of the six is not held out. The checkpoint behind each row is the argmax of that run's AIME24 validation accuracy (Section 11.2), so AIME24 is the set the selection was made on and its column is in-sample: the pre-registration (Appendix C) records that this argmax lifts an arm by about a hundredth in absolute accuracy, and by an amount that differs between arms, so the bias does not cancel cleanly inside a contrast. The column is marked † and kept — dropping a registered endpoint after seeing it is the worse of the two errors — and the five remaining benchmarks, which played no part in selection, carry no such bias and only more variance.

The quantity fixed in advance as the headline here is not any single cell but *direction consistency*: in how many of the six benchmarks the ordering observed on AIME24 is preserved. Four or better of six we declared support, three equivocal, two or fewer evidence against. The observed count is – of six for STEER-F > UNIFORM across the – seeds those two arms share, and – of six for STEER-F >

arm	AIME24†	AIME25	AMC23	MATH500	Minerva	Olympiad	mean
–							

Table 9: Six-benchmark evaluation of each arm's selected checkpoint, averaged over that arm's own seeds. The rightmost column is the unweighted mean over the six. †: AIME24 is the set each checkpoint was selected on (Section 11.2), so that column is in-sample; the other five are held out. As in Table 6, these rows average different seed sets and are not meant to be subtracted — the contrasts below are counted within a seed.

PERMUTED across –. Counting the benchmarks on the 5 that played no part in selection gives – and –: the registered six-count is the decision rule, and the held-out count is printed beside it so a reader can see how much of it survives without the set the checkpoints were chosen on. This is the test that distinguishes a method from an artifact of one question set, and it is reported whatever it returns.

These two contrasts carry the attribution because each changes one thing: UNIFORM has the apparatus without A_H 's value, PERMUTED has its value without its pairing. The differences against GRPO and STEER are registered as well and appear in Table 8, but they change three things and two respectively, so a benchmark difference there names no cause. Arms are paired inside a seed throughout, for the reason Table 6's caption gives; the arm-mean rows of Table 9 are not what is being counted.

The paired error bar. A single measurement of one arm on AIME24's 30 questions carries an across-problem standard error of .0281, which is why the gaps in Table 8 are a fraction of one problem in absolute terms. That figure is the wrong one for a contrast: the two arms answer the same questions, so per-problem difficulty cancels between them and what matters is the standard error of the *paired* difference. This evaluation records one row per generated sample, so on the three avg@1 benchmarks with enough problems to support it — MATH500, OlympiadBench and Minerva, – problems together, over the – seeds the arms share — the paired quantity is computed rather than asserted. STEER-F – UNIFORM is – ± – ($t = -$) against an unpaired error of –, and STEER-F – PERMUTED is – ± – ($t = -$) against –. How much pairing buys is decided by how often the two arms disagree on a problem, which is – and – here; the measurement is what we report, not the reduction we wanted. Two cautions travel with it. At avg@1 each prob-

arm	what changes	acc@32	maj@32
<i>the damping coefficient</i>			
$\lambda = 0.1$	weaker	.1365	.2360
$\lambda = 0.25$	as used	.1491	.2301
$\lambda = 0.5$	stronger	.1444	.2265
<i>the sampler, with the forecast switched off</i>			
$\lambda = 0$, tree	tree only	.1439	.2309
STEER	neither	.1371	.2090
<i>the forecaster, replaced by what the trainer already has</i>			
ORACLE	no heads	.1369	.2234
<i>the base method at its released damping strength</i>			
STEER, $\lambda_{\min}=0.8$	weaker	—	—
<i>relaxed ratio clipping ($\epsilon_{hi}=5$, $\epsilon_{lo}=0.99$)</i>			
STEER-F	—	.1516	.2304
STEER	—	.1393	.2063
<i>other advantage estimators</i>			
RLOO, STEER-F	—	—	—
RLOO, STEER	—	—	—
OPO, STEER-F	—	—	—
OPO, STEER	—	—	—

Table 10: Follow-up ablations, converged window, one seed per row and no error bar. The seed is whichever one the arm was run at, and the release emits it with the row; the ORACLE arm is seed 4, run there so that it shares a seed with the STEER-F and UNIFORM runs the six-benchmark evaluation pairs. The $\lambda = 0.25$ line is the reference and is Table 6’s across-seed mean rather than one run. Each block isolates one element of the design.

lem’s score is a single Bernoulli draw, so these errors carry the sampling noise of one generation per problem on top of the difficulty spread; and pooling problems across seeds treats a (seed, problem) pair as one observation, which ignores the clustering by seed that Section 12.8 shows is real. GSM8K is computed the same way and left out of the pool: at avg@1 it sits near saturation, where a paired difference measures the ceiling rather than the method.

12.10 How much of the design is load-bearing

Table 10 varies one element at a time, at a single seed per row. Every comparison in it is therefore $n = 1$: these rows describe a mechanism, and none of them carries an error bar or licenses an effect size. A row is read against the main arm at *its own* seed, which is not the same number as Table 6’s across-seed mean.

The coefficient. λ scales the visitation term before the two channels are combined in Eq. (23). The value used throughout was not tuned on these runs, and the sweep exists to say whether the result rests on a knife edge. A flat response across 0.1–0.5 would say the method is insensitive to it; a peak at 0.25 with both neighbours below would

say it is a tuned quantity and must be reported as one.

The damping strength, and what it does and does not reach. We measure STEER below GRPO by $-.0077$ over 4 seeds, which the base method’s own results contradict. The explanation is not a difference in the code — we read the released weighting function against ours and it agrees line for line, as do every other value in the two launchers — but in $\lambda_{\min}$, which we set to 0.7 by selecting it on STEER-F while the released script uses 0.8. The baseline is therefore running at a damping strength chosen for the method it is the control for, and one that attenuates half again as hard as its own default. Section 4.4 predicts the sign of that: if the local channel is blind at branch points and anti-correlated away from them, more of it should cost accuracy. This arm is stock STEER ($\lambda = 0$) at the released value, and separates a failure to reproduce from an over-damped control.

The reach of the problem is bounded, and by the design rather than by luck. All four damped arms share $\lambda_{\min} = 0.7$ exactly, so it cancels in any contrast between two of them. STEER-F — UNIFORM and STEER-F — PERMUTED — the pair that carries this paper’s central claim, that the forecast’s *value* is what matters — hold $\lambda_{\min}$, λ , the sampler and the weight map all fixed and vary only what A_H contains. Those two are unaffected by how $\lambda_{\min}$ was chosen. What the choice does reach is the two contrasts against an arm that does not share it: STEER-F — STEER, where only the treatment’s side was tuned, and STEER — GRPO, where GRPO has no such knob at all. We report those two with this stated rather than resolved.

The sampler. This is the cell no other arm supplies. Tree rollouts make a group share prefixes, which changes the GRPO advantage estimate itself, quite apart from any damping. UNIFORM and PERMUTED both run at $\lambda = .25$ and so cannot separate the sampler from the correction. The $\lambda = 0$ tree arm switches the forecast off while leaving the sampler in place, and is the direct answer to “is tree rollout simply a better sampler”. Read against STEER, which has neither, its difference is the sampler’s own contribution; read against STEER-F, what remains is the forecast’s.

The answer is not the one the rest of this paper was written expecting. The sampler alone reaches .1439 against STEER’s .1371, and .2309 against .2090 — about two thirds of STEER-F’s accuracy

margin over STEER and the large majority of its majority-vote margin. Its uplift, .0870, is not below STEER-F's .0810. What remains for the forecast is the accuracy gap between .1439 and .1491.

Two things keep this from being decisive, and neither rescues the original reading. Every cell here is one run, and the spread across this block is about twice the run-to-run spread measured in Section 12.8 — suggestive, not settled. And the λ sweep is not monotone: $\lambda = .1$ sits below $\lambda = 0$ on accuracy, which is what noise looks like at this sample size. We report the block as measured and do not reconcile it. The five-seed campaign of Appendix C decides whether the sampler or the forecast carries the structural term; until then, UNIFORM and PERMUTED establish that the forecast's *value* matters at fixed λ , and nothing in this paper establishes that the tree does not.

The forecaster, replaced by what the trainer already has. Section 12.7 builds the same sibling-differenced quantity from the policy's own next-token entropy and finds it close to A_H . That measurement is taken on a frozen policy at one validation step, and it cannot say what the substitution does *inside* the loop, where the weights it produces move the trajectory the next measurement would be taken on. The ORACLE arm makes the substitution there: same tree, same λ , same weight map, with H_{togo} read from the realised entropy of the rollout verl has already computed. A2 and A3 — the head substitution and the affine calibration — are the only things it removes, and removing them is the whole of its cost difference.

It is cheaper, and by the amount the forward pass costs: 416 s per optimization step against STEER-F's 516 at the same seed, a saving of 19.3%. Both are the window's training steps on the four-GPU box, where every arm is substantially faster than on the two-GPU seeds, so neither is comparable to a step time quoted for a different box.

It is also behind on accuracy. Over the 8 validation points of the converged window, STEER-F at seed 4 leads it by +.0112 on the endpoint this campaign reports and by +.0104 under majority vote. We therefore keep A2 and A3 rather than replacing the forecaster with the realised value, and we state the weight of that decision exactly: one run on each side, one seed. The gap is larger than STEER-F's own range across seeds, .0016, and smaller than GRPO's, .0210. It orders the two arms on this seed; it does not measure the difference between them,

arm	steps	wall clock	acc@32
STEER-F	110	—	.1491
GRPO, step-matched	110	—	.1436
GRPO, compute-matched	—	—	—

Table 11: Equal steps versus equal compute. The matched step is read from the integrated per-step time of the long GRPO run, on the same hardware as its STEER-F reference; times measured on a different machine are not comparable and are not mixed in.

and the measurement that would be this arm at more seeds.

Clipping and the advantage estimator. The relaxed-clipping setting is where entropy interventions are supposed to matter most, because ratio clipping otherwise removes much of the update a token weight would have scaled. The RLOO and OPO rows test whether the construction depends on GRPO specifically: the sibling-prefix baseline of Eq. (15) depends only on the group sharing a prefix, which those estimators also provide, so the method should transfer unchanged. Each is reported beside its own base-method control, so every comparison stays within one estimator.

12.11 Compute-matched comparison

STEER-F costs more per step than GRPO, almost entirely in the MTP forward pass, so an equal-step comparison hands it more wall clock. The honest comparison holds compute fixed instead. We run GRPO to — steps, integrate its measured per-step time, and read its accuracy at the last step whose cumulative time fits inside what STEER-F spent reaching step 110 — rather than assuming a cost ratio.

Both readings are reported. If the gain survives the compute-matched comparison it is a gain; if it does not, the method buys accuracy with time and should be described that way.

12.12 Does the sign survive a change of backbone?

Everything above is one model. A reviewer's first question is whether the effect is a property of the method or of that model, and the honest form of the answer is narrow: on three further backbones we run only GRPO, the base method, and ours, and we ask whether the *ordering* holds. Absolute accuracies are not comparable across rows—the backbones differ in scale, in pre-training family, in whether mathematics was specialized for,

	GRPO	base	ours	Δ GRPO	Δ base
1.5B Qwen-Math	.1436	.1371	.1491	+.0055	+.0120
7B Qwen-Math	-	-	-	-	-
3B Llama	-	-	-	-	-
7B Mistral	-	-	-	-	-

Table 12: The headline contrast on four backbones, seed 1, converged window. The two rightmost columns are within-backbone differences; columns are not averaged, because the rows are not on a common scale. The Llama row uses the instruction-tuned checkpoint (Section 11.2); the other three are base models.

and (Section 11.2) in which benchmark selects the checkpoint—so the table reports each backbone’s own three numbers and the within-backbone differences, and nothing is averaged down a column.

Two questions are separable here, and the design answers them one at a time. The 7B model shares its tokenizer, architecture and pre-training corpus with the 1.5B, so a difference between those two rows is about *scale*. The Llama and Mistral backbones share none of that, so a difference there is about *family*, and in particular about whether the effect needs a base policy that is already sharp on mathematics. Our account predicts that the visitation channel matters wherever the rollout distribution has branch points that differ in what they lead to, which is a property of the task rather than of the pre-training corpus; a backbone on which the sign reverses would be evidence against that account, not merely a weaker result.

The statistic we report is the sign of Δ_{GRPO} , and how many of the four backbones share it. With one seed per cell, a single backbone’s difference carries about the same uncertainty as a single seed on the primary backbone (Section 12.8), so no individual row is decisive; the claim rests on agreement across rows. We report the count, $-$ of 4, and treat a split result as a limit on generality rather than as noise to be averaged away.

12.13 Cost

Within the runs executed on identical hardware, STEER takes 833s per optimization step and STEER-F takes 1516s, an overhead of 82%. The UNIFORM and PERMUTED arms cost 1480 and 1492s, confirming that the cost is the tree rollout plus the MTP forward pass and not the A_H computation, which is a subtraction over a mask. We do not report a GRPO timing because that arm ran on a different machine.

12.14 A note on how these numbers were selected

We report window means rather than the best checkpoint, and the difference is not cosmetic. Selecting the maximum over the twelve validation points inflates every arm by a similar amount: $+.013$ for STEER, $+.011$ for UNIFORM, $+.016$ for PERMUTED and $+.017$ for STEER-F, relative to the window mean. Since a single AIME24 measurement carries a standard error of $.0281$ across problems, a maximum over a dozen noisy points is a biased statistic whose bias grows with the number of checkpoints compared. The arm ordering happens to be the same under both rules here, but we regard the window mean as the defensible one and use it throughout.

13 Conclusion

The gradient of trajectory entropy has two terms, and the token-level entropy controls now in use estimate one of them. We showed that the omission is structural: the estimated term vanishes exactly where the omitted term is largest, and it vanishes for a correct reason, so the gap cannot be closed by improving the estimator. We then showed that the omitted term is a REINFORCE estimator whose reward-to-go is the remaining conditional entropy, and that a tree-structured rollout supplies the sibling baseline it requires — which is what makes the correction estimable at all. The resulting correction is mean-preserving by construction: it does not slow entropy collapse, it redirects it. The campaign measures that consequence and finds it. The aggregate entropy moves when the damping apparatus is switched and does not move detectably when only A_H ’s contents change, while accuracy separates the same two groups in the opposite direction. On the seeds in hand that is a statement about what is detectable, not a proof that the aggregate is untouched.

Estimating H_{togo} is a separate problem, and the decomposition does not depend on how it is solved. Ours is a parallel multi-token-prediction forecaster whose error has a known sign and monotonicity in closed form; we also measure it to reproduce a quantity the trainer already computes at $R^2 = 0.840$, and to cost the step time reported in Section 12.13 to do so. That is a reason to look for a cheaper estimator of this reward-to-go, not a reason to doubt the channel: the head-free ORACLE arm of Table 10 is the control that separates those

two readings, and it has been run. At the one seed it carries, the realised value costs 19.3% less per step and reads $+0.0112$ below the forecast on the primary endpoint. That is a reason to keep the estimator rather than a licence to declare it necessary: a single draw on each side orders the two and does not size the difference.

Limitations

A pre-registered falsification condition is met. Appendix C fixed, before these runs, that if STEER-F – UNIFORM straddled zero across seeds then the forecast’s *value* does not matter and the apparatus explains the effect. On the registered primary endpoint it straddles: -0.0020 to $+0.0182$ over 3 paired seeds, with 1 negative. That clause calls this the claim of the paper, and we do not renegotiate it here. What it does not reach is the companion condition: STEER-F – PERMUTED, which destroys only the pairing of A_H to a branch and keeps its magnitudes, does not straddle (0 of 3 negative), nor do the two structural conditions on the entropy ordering and on the mean token weight. The honest summary is that the pairing survives its registered test and the value does not, at this seed count, on this benchmark. Section 12.4 states the tension that leaves between the pre-registration’s choice of central claim and the one this paper leads with.

The forecaster does not earn its cost. The calibration in Eq. (19) suppresses heads beyond the first, so $\hat{H}_{\text{togo}}$ is dominated by a single step ahead. This is the predicted behavior of Eq. (18) rather than a bug, but Section 12.7 measures the consequence and it is worse than a short horizon: put through the same sibling baseline, the policy’s own next-token entropy reproduces A_H at $r = +0.917$ and $R^2 = 0.840$, with the magnitudes in the ratio 0.741 that γ_{Ha_1} predicts. On that evidence the MTP heads, their warm-up and their calibration are machinery around a quantity the trainer already computes, and they carry the overhead of Section 12.13 to do it. The ORACLE arm of Table 10 is the direct test and it does not agree: made inside the training loop, the substitution saves 19.3% of the step time and gives up $+0.0112$ on the primary endpoint at the one seed it was run at. Both readings stand as measured. A correlation on a frozen policy does not predict a training loop, and one draw on each side does not overturn a correlation over 4 prompts; whether the heads earn their cost is left open here, and the arm that closes it is the same

arm at more seeds. What the measurement does not reach is the visitation channel itself: the local twin is not Ω , and it exists only because the tree supplies siblings to difference against. A forecaster whose error did not grow with k would test the idea more sharply; a sequential forecaster is the obvious comparison and we have not run it.

Few seeds, and a headline gap inside the spread.

Every arm has now finished more than once, so the contrasts here do carry a seed-level test — but on 3 to 4 paired seeds depending on the row, which is few. This bounds the effect sizes rather than the structural result: the dissociation of Section 12.3 is a comparison between two groups of contrasts whose $|t|$ do not overlap, and no single row carries it. The accuracy gap is the claim that does rest on one row, and Section 12.8 shows the floor under it is not yet measured — GRPO’s converged-window accuracy moves $.0210$ across its 4 seeds, STEER-F’s $.0016$ across 3, and PERMUTED’s only $.0095$, three estimates of the same noise differing by more than an order of magnitude, so none of them bounds the others. Against the largest, the headline $+0.0055$ is the smaller number. Rollout sampling is not seeded across processes, so re-running a configuration is a second draw rather than a restoration — which is what makes the repeat measurable and what makes it necessary. The pre-registered five-seed campaign of Appendix C is what would settle the size; what this paper settles without it is the structural result and the ordering of the arms.

A single validation benchmark, and what our t -statistics do not cover. Every claim rests on AIME 2024, which has 30 problems. A single measurement of one arm carries an across-problem standard error of $.0281$, so the gaps in Table 8 are a fraction of one problem in absolute terms. The paired t in that table is taken over seeds, so it speaks to re-running a configuration and not to a different set of questions; the within-run version, over the eight validation steps of a single trajectory, is reported separately in Appendix D so the two are not read as one. The statistic that would speak to generalization — the across-problem standard error of the *paired* difference, in which per-problem difficulty cancels between two arms evaluated on the same questions — is not recoverable from the *training* logs, which record per-benchmark means and standard deviations rather than per-problem scores. The six-benchmark evaluation records the per-problem scores and so does carry it, on the

three benchmarks large enough to support one; those figures are in Section 12.9. What remains missing is the same statistic for AIME24 itself, which is where every trajectory-level number in this paper is measured, and which is also the set the checkpoints were selected on. The gap this paragraph names is narrowed, not closed.

One prediction of the analysis is unmeasured. Section 4.4 argues that the two channels are anti-correlated. The prefactor half of that claim is analytic and the A_H half is inferred from a correlation with branch diversity; we have not measured their *product* at real sibling-divergence positions in a live rollout. Until that measurement exists, Table 1 is a prediction rather than an observation.

A second one is owed by the result we lead with. Section 9.3 says a sum-zero correction should widen the spread *across siblings* at the positions it reaches, which is the positive claim behind the dissociation of Section 12.3; the release instruments the support-restricted split that tests it, and no run reported here carries it yet. The dissociation we do measure is the negative half — that the aggregate does not move with A_H — and the two are not the same statement.

The support is small by construction. Section 9 bounds the fraction of positions carrying a non-zero A_H at $2(n - 1)/T$, on the order of a percent for our settings. Whether the effect scales with n , or with a branching schedule chosen adaptively rather than at fixed depths, is untested.

One hyperparameter was selected on the method and applied to its baseline. $\lambda_{\min}$, the floor of the weight map, is 0.7 throughout, chosen because it was best for STEER-F; the base method’s released script uses 0.8. The base method therefore appears here at a damping strength picked for the method it controls for, and at 1.5 times the maximum attenuation of its own default, which is the most likely reason we measure it below GRPO where its authors do not. We did not re-tune $\lambda_{\min}$ per arm, and a reader who wants each method at its own best setting will not find it here.

What this does not reach is the claim the paper makes. All four damped arms carry the same $\lambda_{\min}$, so it cancels between any two of them, and the contrasts that separate the forecast’s value from its apparatus — STEER-F against UNIFORM and against PERMUTED — hold it fixed along with everything else. The two contrasts it does reach, STEER-F

— STEER and STEER — GRPO, are reported with the confound stated; Section 12.10 adds the base method at its released value so the reproduction question can be read on its own.

An assumption with no remedy. Approximation B2 — that a token’s update does not change the conditional distribution at other positions — is false in a network with shared parameters. It is inherited from the base method rather than introduced here, but our construction adds a second estimator that relies on it.

Scope of the backbone evidence. The full decomposition—the five arms, the ablations, the mechanism diagnostics—rests on one 1.5B model. The three further backbones (Section 12.12) carry the three-way contrast at one seed each, which establishes the sign and not its size, and they stop at 7B: we have no evidence about models an order of magnitude larger, where the base policy is sharper and the relative size of the two channels may differ. Family and scale are also not fully crossed—the mathematics backbones span 1.5B to 7B and the others sit at 3B and 7B—so what the latter test is whether the sign survives a different pre-training corpus, not whether it survives one at 8B or beyond. Those rows further differ from the mathematics ones in more than family. They are selected on a different benchmark (Section 11.2), and the Llama row is instruction-tuned where the other three are base models, which was forced rather than chosen: the base checkpoint left almost every training group degenerate. Each difference is what makes the row measurable at all, and each is a reason no claim here compares accuracy across backbones—only the sign of a within-backbone contrast is read.

Ethics Statement

This work studies an optimization-time regularizer for reinforcement learning on mathematics problems with automatically verifiable answers. It introduces no new data collection, no human subjects, and no new model release. The intervention acts on which of several equally-scored continuations receives a slightly smaller gradient weight; it does not alter the reward, and so does not change what behavior is being optimized for. The training data is a public competition-mathematics set and the evaluation set is a public examination.

The main foreseeable risk is indirect. Methods that preserve diversity during RL fine-tuning are

not specific to mathematics, and the same mechanism would apply to a reward model encoding any objective, including a harmful one; we note that our method inherits, and does not mitigate, whatever the reward function encodes. The computational cost of the method (a forward pass through a set of small prediction heads at every optimization step) is a real environmental cost, which we quantify in Section 12.13 so that it can be weighed against the benefit.

References

Arash Ahmadian, Chris Cremer, Matthias Gallé, Marzieh Fadaee, Julia Kreutzer, Olivier Pietquin, Ahmet Üstün, , and Sara Hooker. 2024. Back to basics: Revisiting reinforce-style optimization for learning from human feedback in llms. In Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 12248–12267.

Shengnan An, Xunliang Cai, Xuezhi Cao, Xiaoyu Li, Yehao Lin, Junlin Liu, Xinxuan Lv, Dan Ma, Xuanlin Wang, Ziwen Wang, , et al. 2025. Amo-bench: Large language models still struggle in high school math competitions. ArXiv preprint arXiv:2510.26768.

Anonymous. 2025. Rethinking entropy regularization in large reasoning models. ArXiv preprint arXiv:2509.25133.

Anthropic. 2025. Introducing claude opus 4.5. <https://www.anthropic.com/news/claude-opus-4-5>.

Sikai Bai, Haoxi Li, Jie Zhang, Yongjiang Liu, , and Song Guo. 2026. Ttvs: Boosting self-exploring reinforcement learning via test-time variational synthesis. ArXiv preprint arXiv:2604.08468.

Tianle Cai, Yuhong Li, Zhengyang Geng, Hongwei Peng, Jason D. Lee, Deming Chen, and Tri Dao. 2024. Medusa: Simple LLM inference acceleration framework with multiple decoding heads. ArXiv preprint arXiv:2401.10774.

Yupeng Chang, Yi Chang, , and Yuan Wu. 2026. Balora: Bias-alleviating low-rank adaptation to mitigate catastrophic inheritance in large language models. In The Fourteenth International Conference on Learning Representations.

Yupeng Chang, Chenlu Guo, Yi Chang, , and Yuan Wu. 2025. Lora-mgpo: Mitigating double descent in low-rank adaptation via momentum-guided perturbation optimization. in findings of the association for computational linguistics: Emnlp 2025, pages 648–659.

Yupeng Chang, Xu Wang, Jindong Wang, Yuan Wu, Linyi Yang, Kaijie Zhu, Hao Chen, Xiaoyuan Yi, Cunxiang Wang, Yidong Wang, , et al. 2024. A survey on evaluation of large language models. acm

transactions on intelligent systems and technology, 15(3):1–45.

Jiawei Chen, Hande Dong, Xiang Wang, Fuli Feng, Meng Wang, , and Xiangnan He. 2023. Bias and debias in recommender system: A survey and future directions. acm transactions on information systems, 41(3):1–39.

Daixuan Cheng, Shaohan Huang, Xuekai Zhu, Bo Dai, Wayne Xin Zhao, Zhenliang Zhang, , and Furu Wei. 2025. Reasoning with exploration: An entropy perspective. ArXiv preprint arXiv:2506.14758.

Xiangxiang Chu, Hailang Huang, Xiao Zhang, Fei Wei, , and Yong Wang. 2025. Gpg: A simple and strong reinforcement learning baseline for model reasoning. ArXiv preprint arXiv:2504.02546.

Ganqu Cui, Lifan Yuan, Zefan Wang, Hanbin Wang, Wendi Li, Bingxiang He, Yuchen Fan, Tianyu Yu, Qixin Xu, Weize Chen, , et al. 2025a. Process reinforcement through implicit rewards. ArXiv preprint arXiv:2502.01456.

Ganqu Cui, Yuchen Zhang, Jiacheng Chen, Lifan Yuan, Zhi Wang, Yuxin Zuo, Haozhan Li, Yuchen Fan, Huayu Chen, Weize Chen, , et al. 2025b. The entropy mechanism of reinforcement learning for reasoning language models. ArXiv preprint arXiv:2505.22617.

Yu Cui, Feng Liu, Jiawei Chen, Canghong Jin, Xingyu Lou, Changwang Zhang, Jun Wang, Yuegang Sun, , and Can Wang. 2025c. Hatllm: Hierarchical attention masking for enhanced collaborative modeling in llm-based recommendation. ArXiv preprint arXiv:2510.10955.

Jia Deng, Jie Chen, Zhipeng Chen, Wayne Xin Zhao, , and Ji-Rong Wen. 2025. Decomposing the entropy-performance exchange: The missing keys to unlocking effective reinforcement learning. ArXiv preprint arXiv:2508.02260.

Chongming Gao, Kexin Huang, Jiawei Chen, Yuan Zhang, Biao Li, Peng Jiang, Shiqi Wang, Zhong Zhang, , and Xiangnan He. 2023a. Alleviating matthew effect of offline reinforcement learning in interactive recommendation. In Proceedings of the 46th international ACM SIGIR conference on research and development in information retrieval, pages 238–248.

Chongming Gao, Shiqi Wang, Shijun Li, Jiawei Chen, Xiangnan He, Wenqiang Lei, Biao Li, Yuan Zhang, , and Peng Jiang. 2023b. Cirs: Bursting filter bubbles by counterfactual interactive recommender system. acm transactions on information systems, 42(1):1–27.

Google. 2025. A new era of intelligence with gemini 3. <https://blog.google/products/gemini/gemini-3/>.

Aaron Grattafiori, Abhimanyu Dubey, Abhinav Jauhri, Abhinav Pandey, Abhishek Kadian, Ahmad Al-Dahle, Aiesha Letman, Akhil Mathur, Alan Schelten,

1996	Alex Vaughan, , et al. 2024. The llama 3 herd of models. ArXiv preprint arXiv:2407.21783.	Binyuan Hui, Jian Yang, Zeyu Cui, Jiayi Yang, Dayiheng Liu, Lei Zhang, Tianyu Liu, Jiajun Zhang, Bowen Yu, Keming Lu, , et al. 2024. Qwen2.5-coder technical report. ArXiv preprint arXiv:2409.12186.	2048
1997			2049
1998	Aaron Grattafiori et al. 2024. The llama 3 herd of models. ArXiv preprint arXiv:2407.21783.		2050
1999			2051
2000	Daya Guo, Dejian Yang, Haowei Zhang, Junxiao Song, Ruoyu Zhang, Runxin Xu, Qihao Zhu, Shirong Ma, Peiyi Wang, Xiao Bi, , et al. 2025. Deepseek-r1: Incentivizing reasoning capability in llms via reinforcement learning. ArXiv preprint arXiv:2501.12948.	Zed Industries. 2025. zeta. https://huggingface.co/datasets/zed-industries/zeta .	2052
2001			2053
2002		Aaron Jaech, Adam Kalai, Adam Lerer, Adam Richardson, Ahmed El-Kishky, Aiden Low, Alec Helyar, Aleksander Madry, Alex Beutel, Alex Carney, , et al. 2024. Openai o1 system card. ArXiv preprint arXiv:2412.16720.	2054
2003			2055
2004			2056
2005	Tuomas Haarnoja, Aurick Zhou, Pieter Abbeel, , and Sergey Levine. 2018. Soft actor-critic: Off-policy maximum entropy deep reinforcement learning with a stochastic actor. in international conference on machine learning, pages 1861–1870. pmlr.		2057
2006			2058
2007		Naman Jain, King Han, Alex Gu, Wen-Ding Li, Fanjia Yan, Tianjun Zhang, Sida Wang, Armando Solar-Lezama, Koushik Sen, , and Ion Stoica. 2024. Livecodebench: Holistic and contamination free evaluation of large language models for code. ArXiv preprint arXiv:2403.07974.	2059
2008			2060
2009			2061
2010	Yaru Hao, Li Dong, Xun Wu, Shaohan Huang, Zewen Chi, , and Furu Wei. 2025. On-policy rl with optimal reward baseline. ArXiv preprint arXiv:2505.23585.		2062
2011			2063
2012			2064
2013	Zhezheng Hao, Hong Wang, Haoyang Liu, Jian Luo, Jiarui Yu, Hande Dong, Qiang Lin, Can Wang, and Jiawei Chen. 2026. Rethinking entropy interventions in RLVR: An entropy change perspective. In <i>Proceedings of the 64th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)</i> , pages 31105–31133.	Albert Q. Jiang et al. 2023. Mistral 7b. ArXiv preprint arXiv:2310.06825.	2065
2014			2066
2015		Yihang Jiang, Xiaoyang Li, Guangxu Zhu, Hang Li, Jing Deng, Kaifeng Han, Chao Shen, Qingjiang Shi, , and Rui Zhang. 2023. 6g non-terrestrial networks enabled low-altitude economy: Opportunities and challenges. ArXiv preprint arXiv:2311.09047.	2067
2016			2068
2017			2069
2018			2070
2019			2071
2020	Andre He, Daniel Fried, , and Sean Welleck. 2025a. Rewarding the unlikely: Lifting grpo beyond distribution sharpening. ArXiv preprint arXiv:2506.02355.	Nathan Lambert, Jacob Morrison, Valentina Pyatkin, Shengyi Huang, Hamish Ivison, Faeze Brahman, Lester James V Miranda, Alisa Liu, Nouha Dziri, Shane Lyu, , et al. 2024. Tulu 3: Pushing frontiers in open language model post-training. ArXiv preprint arXiv:2411.15124.	2072
2021			2073
2022			2074
2023	Chaoqun He, Renjie Luo, Yuzhuo Bai, Shengding Hu, Zhen Leng Thai, Junhao Shen, Jinyi Hu, Xu Han, Yujie Huang, Yuxiang Zhang, , et al. 2024. Olympiad-bench: A challenging benchmark for promoting agi with olympiad-level bilingual multimodal scientific problems. ArXiv preprint arXiv:2402.14008.	Aitor Lewkowycz, Anders Andreassen, David Dohan, Ethan Dyer, Henryk Michalewski, Vinay Ramasesh, Ambrose Slone, Cem Anil, Imanol Schlag, Theo Gutman-Solo, , et al. 2022. Solving quantitative reasoning problems with language models. advances in neural information processing systems, 35:3843–3857.	2075
2024			2076
2025			2077
2026			2078
2027			2079
2028			2080
2029	Jujie He, Jiakai Liu, Chris Yuhao Liu, Rui Yan, Chaojie Wang, Peng Cheng, Xiaoyu Zhang, Fuxiang Zhang, Jiacheng Xu, Wei Shen, , et al. 2025b. Skywork open reasoner 1 technical report. ArXiv preprint arXiv:2505.22312.		2081
2030			2082
2031			2083
2032			2084
2033		Jia Li, Edward Beeching, Lewis Tunstall, Ben Lipkin, Roman Soletskyi, Shengyi Huang, Kashif Rasul, Longhui Yu, Albert Q Jiang, Ziju Shen, , et al. 2024. Numinamath: The largest public dataset in ai4maths with 860k pairs of competition math problems and solutions. hugging face repository, 13(9):9.	2085
2034	Dan Hendrycks, Collin Burns, Saurav Kadavath, Akul Arora, Steven Basart, Eric Tang, Dawn Song, , and Jacob Steinhardt. 2021. Measuring mathematical problem solving with the math dataset. ArXiv preprint arXiv:2103.03874.		2086
2035			2087
2036			2088
2037			2089
2038			2090
2039	Jian Hu, Jason Klein Liu, Haotian Xu, , and Wei Shen. 2025a. Reinforce++: An efficient rlhf algorithm with robustness to both prompt and reward models. ArXiv preprint arXiv:2501.03262, 1(3):5.	Mengdi Li, Jiaye Lin, Xufeng Zhao, Wenhao Lu, Peilin Zhao, Stefan Wermter, , and Di Wang. 2025a. Curriculum-rlaif: Curriculum alignment with reinforcement learning from ai feedback. ArXiv preprint arXiv:2505.20075.	2091
2040			2092
2041			2093
2042			2094
2043	Jingcheng Hu, Yinmin Zhang, Qi Han, Daxin Jiang, Xiangyu Zhang, , and Heung-Yeung Shum. 2025b. Open-reasoner-zero: An open source approach to scaling up reinforcement learning on the base model. ArXiv preprint arXiv:2503.24290.	Qingbin Li, Rongkun Xue, Jie Wang, Ming Zhou, Zhi Li, Xiaofeng Ji, Yongqi Wang, Miao Liu, Zheming Yang, Minghui Qiu, , et al. 2025b. Cure: Critical-token-guided re-concatenation for entropy-collapse prevention. ArXiv preprint arXiv:2508.11016.	2095
2044			2096
2045			2097
2046			2098
2047			2099
			2100

2101	Siyi Lin, Chongming Gao, Jiawei Chen, Sheng Zhou,	Yuda Song, Julia Kempe, , and Remi Munos. 2025.	2158
2102	Binbin Hu, Yan Feng, Chun Chen, , and Can Wang.	Outcome-based exploration for llm reasoning. ArXiv	2159
2103	2025. How do recommendation models amplify pop-	preprint arXiv:2509.06941.	2160
2104	ularity bias? an analysis from the spectral perspective.		
2105	In Proceedings of the Eighteenth ACM International	Hongze Tan and Jianfei Pan. 2025. Gtpo and grpo-s:	2161
2106	Conference on Web Search and Data Mining, pages	Token and sequence-level reward shaping with policy	2162
2107	659–668.	entropy. ArXiv preprint arXiv:2508.04349.	2163
2108	Aixin Liu, Aoxue Mei, Bangcai Lin, Bing Xue, Bingx-	Haozhe Wang, Qixin Xu, Che Liu, Junhong Wu,	2164
2109	uan Wang, Bingzheng Xu, Bochao Wu, Bowei Zhang,	Fangzhen Lin, , and Wenhui Chen. 2025a. Emergent	2165
2110	Chaofan Lin, Chen Dong, , et al. 2025a. Deepseek-	hierarchical reasoning in llms through reinforcement	2166
2111	v3. 2: Pushing the frontier of open large language	learning. ArXiv preprint arXiv:2509.03646.	2167
2112	models. ArXiv preprint arXiv:2512.02556.		
2113	Mingjie Liu, Shizhe Diao, Ximing Lu, Jian Hu, Xin	Hong Wang, Zhezheng Hao, Jian Luo, Chenxing Wei,	2168
2114	Dong, Yejin Choi, Jan Kautz, , and Yi Dong. 2025b.	Yao Shu, Lei Liu, Qiang Lin, Hande Dong, , and	2169
2115	Prorl: Prolonged reinforcement learning expands rea-	Jiawei Chen. 2025b. Scheduling your llm reinforce-	2170
2116	soning boundaries in large language models. ArXiv	ment learning with reasoning trees. ArXiv preprint	2171
2117	preprint arXiv:2505.24864.	arXiv:2510.24832.	2172
2118	Ziyu Liu, Yuhang Zang, Shengyuan Ding, Yuhang	Jiakang Wang, Runze Liu, Fuzheng Zhang, Xiu Li, ,	2173
2119	Cao, Xiaoyi Dong, Haodong Duan, Dahua Lin, ,	and Guorui Zhou. 2025c. Stabilizing knowledge,	2174
2120	and Jiaqi Wang. 2025c. Spark: Synergistic policy	promoting reasoning: Dual-token constraints for rlvr.	2175
2121	and reward co-evolving framework. ArXiv preprint	ArXiv preprint arXiv:2507.15778.	2176
2122	arXiv:2509.22624.		
2123	Michael Luo, Sijun Tan, Justin Wong, Xiaoxiang Shi,	Shenzhi Wang, Le Yu, Chang Gao, Chujie Zheng, Shix-	2177
2124	William Y Tang, Manan Roongta, Colin Cai, Jeffrey	uan Liu, Rui Lu, Kai Dang, Xionghui Chen, Jianxin	2178
2125	Luo, Tianjun Zhang, Li Erran Li, , et al. 2025. Deep-	Yang, Zhenru Zhang, , et al. 2025d. Beyond the	2179
2126	scaler: Surpassing o1-preview with a 1.5 b model by	80/20 rule: High-entropy minority tokens drive effec-	2180
2127	scaling rl. notion blog.	tive reinforcement learning for llm reasoning. ArXiv	2181
		preprint arXiv:2506.01939.	2182
2128	Shichao Ma, Zhiyuan Ma, Ming Yang, Xiaofan Li, Xing	Ziqing Wang, Yibo Wen, William Pattie, Xiao Luo,	2183
2129	Wu, Jintao Du, Yu Cheng, Weiqiang Wang, Qiliang	Weimin Wu, Jerry Yao-Chieh Hu, Abhishek Pandey,	2184
2130	Liu, Zhengyang Zhou, , et al. 2026. Tspo: Break-	Han Liu, , and Kaize Ding. 2025e. Polo: Pref-	2185
2131	ing the double homogenization dilemma in multi-	erenceguided multi-turn reinforcement learning for	2186
2132	turn search policy optimization. ArXiv preprint	lead optimization. ArXiv preprint arXiv:2509.21737.	2187
2133	arXiv:2601.22776.		
2134	Volodymyr Mnih, Adria Puigdomenech Badia, Mehdi	Fang Wu, Weihao Xuan, Ximing Lu, Zaid Harchaoui, ,	2188
2135	Mirza, Alex Graves, Timothy Lillicrap, Tim Harley,	and Yejin Choi. 2025a. The invisible leash: Why	2189
2136	David Silver, , and Koray Kavukcuoglu. 2016. Asyn-	rlvr may not escape its origin. ArXiv preprint	2190
2137	chronous methods for deep reinforcement learning. in	arXiv:2507.14843.	2191
2138	international conference on machine learning, pages	Fei Wu, Zhenrong Zhang, Qikai Chang, Jianshu Zhang,	2192
2139	1928–1937. pmlr.	Quan Liu, , and Jun Du. 2026. Step potential advan-	2193
2140	OpenAI. 2025. Introducing gpt-5.2. https://openai.com/index/introducing-gpt-5-2 .	tage estimation: Harnessing intermediate confidence	2194
2141		and correctness for efficient mathematical reasoning.	2195
2142	John Schulman, Filip Wolski, Prafulla Dhariwal, Alec	ArXiv preprint arXiv:2601.03823.	2196
2143	Radford, , and Oleg Klimov. 2017. Proximal	Junkang Wu, Kexin Huang, Jiancan Wu, An Zhang,	2197
2144	policy optimization algorithms. ArXiv preprint	Xiang Wang, , and Xiangnan He. 2025b. Quantile ad-	2198
2145	arXiv:1707.06347.	vantage estimation for entropy-safe reasoning. ArXiv	2199
2146	Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu,	preprint arXiv:2509.22611.	2200
2147	Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan	Zhenhe Wu, Jian Yang, Jiaheng Liu, Xianjie Wu,	2201
2148	Zhang, YK Li, Yang Wu, , et al. 2024. Deepseek-	Changzai Pan, Jie Zhang, Yu Zhao, Shuangy-	2202
2149	math: Pushing the limits of mathematical reason-	ong Song, Yongxiang Li, , and Zhoujun Li.	2203
2150	ing in open language models. ArXiv preprint	2025c. Tablerl: Region-based reinforcement	2204
2151	arXiv:2402.03300.	learning for table understanding. ArXiv preprint	2205
2152	Guangming Sheng, Chi Zhang, Zilingfeng Ye, Xibin	arXiv:2505.12415.	2206
2153	Wu, Wang Zhang, Ru Zhang, Yanghua Peng, Haibin	Can Xie, Ruotong Pan, Xiangyu Wu, Yunfei Zhang,	2207
2154	Lin, , and Chuan Wu. 2025. Hybridflow: A flexi-	Jiayi Fu, Tingting Gao, , and Guorui Zhou. 2025.	2208
2155	ble and efficient rlhf framework. In Proceedings of	Unlocking exploration in rlvr: Uncertainty-aware ad-	2209
2156	the Twentieth European Conference on Computer	vantage shaping for deeper reasoning. ArXiv preprint	2210
2157	Systems, pages 1279–1297.	arXiv:2510.10649.	2211

2212	An Yang, Beichen Zhang, Binyuan Hui, Bofei Gao,	Wenyuan Zhang, Xinghua Zhang, Haiyang Yu, Shuaiyi	2268
2213	Bowen Yu, Chengpeng Li, Dayiheng Liu, Jian-	Nie, Bingli Wu, Juwei Yue, Tingwen Liu, , and Yong-	2269
2214	hong Tu, Jingren Zhou, Junyang Lin, , et al. 2024a.	bin Li. 2026b. Expseek: Self-triggered experience	2270
2215	Qwen2.5-math technical report: Toward mathemat-	seeking for web agents. Preprint, arXiv:2601.08605.	2271
2216	ical expert model via self-improvement. ArXiv		
2217	preprint arXiv:2409.12122.		
2218	Shihui Yang, Chengfeng Dou, Peidong Guo, Kai	Xingjian Zhang, Siwei Wen, Wenjun Wu, , and Lei	2272
2219	Lu, Qiang Ju, Fei Deng, , and Rihui Xin. 2025a.	Huang. 2025d. Edge-grpo: Entropy-driven grpo	2273
2220	Depo: Dynamic clipping policy optimization. ArXiv	with guided error correction for advantage diversity.	2274
2221	preprint arXiv:2509.02333.	ArXiv preprint arXiv:2507.21848.	2275
2222	Weiqin Yang, Jiawei Chen, Xin Xin, Sheng Zhou, Bin-	Xinglang Zhang, Yunyao Zhang, ZeLiang Chen, Jun-	2276
2223	bin Hu, Yan Feng, Chun Chen, , and Can Wang.	qing Yu, Wei Yang, , and Zikai Song. 2026c. Logical	2277
2224	2024b. Psl: Rethinking and improving softmax	phase transitions: Understanding collapse in llm log-	2278
2225	loss from pairwise perspective for recommendation.	ical reasoning. Preprint, arXiv:2601.02902.	2279
2226	advances in neural information processing systems,		
2227	37:120974–121006.		
2228	Weiqin Yang, Jiawei Chen, Shengjia Zhang, Peng Wu,	Yunyao Zhang, Xinglang Zhang, Junxi Sheng, Wen-	2280
2229	Yuegang Sun, Yan Feng, Chun Chen, , and Can Wang.	bing Li, Junqing Yu, Yi-Ping Phoebe Chen, Wei	2281
2230	2025b. Breaking the top-k barrier: Advancing top-k	Yang, , and Zikai Song. 2026d. Semantic-aware log-	2282
2231	ranking metrics optimization in recommender sys-	ical reasoning via a semiotic framework. Preprint,	2283
2232	tems. In Proceedings of the 31st ACM SIGKDD	arXiv:2509.24765.	2284
2233	Conference on Knowledge Discovery and Data Min-		
2234	ing V. 2, pages 3542–3552.	Ziqi Zhao, Zhaochun Ren, Jiahong Zou, Liu Yang, Zhi-	2285
2235	Weiqin Yang, Bohao Wang, Zhenxiang Xu, Jiawei Chen,	wei Xu, Xuri Ge, Zhumin Chen, Xinyu Ma, Daiting	2286
2236	Shengjia Zhang, Jingbang Chen, Canghong Jin, , and	Shi, Shuaiqiang Wang, , et al. 2026. Reinforced effi-	2287
2237	Can Wang. 2026. Bear: Towards beam-searchaware	cient reasoning via semantically diverse exploration.	2288
2238	optimization for recommendation with large lan-	ArXiv preprint arXiv:2601.05053.	2289
2239	guage models. ArXiv preprint arXiv:2601.22925.		
2240	Qiyang Yu, Zheng Zhang, Ruofei Zhu, Yufeng Yuan,	Yixiao Zhou, Yang Li, Dongzhou Cheng, Hehe Fan,	2290
2241	Xiaochen Zuo, Yu Yue, Weinan Dai, Tiantian Fan,	, and Yu Cheng. 2026. Look inward to explore	2291
2242	Gaohong Liu, Lingjun Liu, , et al. 2025. Dapo: An	outward: Learning temperature policy from llm in-	2292
2243	open-source llm reinforcement learning system at	ternal states via hierarchical rl. ArXiv preprint	2293
2244	scale. ArXiv preprint arXiv:2503.14476.	arXiv:2602.13035.	2294
2245	Weihao Zeng, Yuzhen Huang, Qian Liu, Wei Liu, Ke-	Xinyu Zhu, Mengzhou Xia, Zhepei Wei, Wei-Lin Chen,	2295
2246	qing He, Zejun Ma, , and Junxian He. 2025. Simpler-	Danqi Chen, , and Yu Meng. 2025. The surprising	2296
2247	lzo: Investigating and taming zero reinforcement	effectiveness of negative reinforcement in llm reason-	2297
2248	learning for open base models in the wild. ArXiv	ing. ArXiv preprint arXiv:2506.01347.	2298
2249	preprint arXiv:2503.18892.		
2250	Kaiyan Zhang, Yuxin Zuo, Bingxiang He, Youbang	Brian D Ziebart, Andrew Maas, J Andrew Bagnell, ,	2299
2251	Sun, Runze Liu, Che Jiang, Yuchen Fan, Kai Tian,	and Anind K Dey. 2008. Maximum entropy inverse	2300
2252	Guoli Jia, Pengfei Li, , et al. 2025a. A survey of	reinforcement learning. In AAAI, volume 8, pages	2301
2253	reinforcement learning for large reasoning models.	1433–1438.	2302
2254	ArXiv preprint arXiv:2509.08827.		
2255	Kangning Zhang, Wenxiang Jiao, Kounianhua Du,	A Closed Forms for Section 4.4	2303
2256	Yuan Lu, Weiwen Liu, Weinan Zhang, , and Yong		
2257	Yu. 2025b. Looptool: Closing the data-training	Perturbing one logit of a k -way uniform distri-	2304
2258	loop for robust llm tool calls. ArXiv preprint	bution by δ and writing $S := e^\delta + k - 1$ gives	2305
2259	arXiv:2511.09148.	$\pi_1 = e^\delta/S$ and $\pi_j = 1/S$ for $j \neq 1$, so	2306
2260	Ruipeng Zhang, Ya-Chien Chang, , and Sicun Gao.		
2261	2025c. When maximum entropy misleads policy	$H(\delta) = \log S - \frac{\delta e^\delta}{S},$	2307
2262	optimization. ArXiv preprint arXiv:2506.05615.	$H'(\delta) = -\frac{(k-1)\delta e^\delta}{S^2},$	
2263	Shengjia Zhang, Weiqin Yang, Jiawei Chen, Peng	whence $H'(0) = 0$ and $H''(0) = -(k-1)/k^2$.	2308
2264	Wu, Yuegang Sun, Gang Wang, Qihao Shi, , and	The sign of H' is $-\text{sign}(\delta)$, so any perturbation	2309
2265	Can Wang. 2026a. Talos: Optimizing top-k ac-	lowers the entropy, but only at second order. Ta-	2310
2266	curacy in recommender systems. ArXiv preprint	ble 13 contrasts this with a peaked distribution,	2311
2267	arXiv:2601.19276.	where the first-order term does not vanish.	2312

distribution	H_0	Ω	ΔH at $\delta=.1$
uniform, $k=10$	2.303	0	-4.7×10^{-4}
uniform, $k=50$	3.912	0	-1.0×10^{-4}
top .5 + 9 uniform	1.792	-0.549	-5.6×10^{-2}
top .9 + 9 uniform	0.545	-0.396	-3.8×10^{-2}

Table 13: First-order prediction versus actual entropy change. At a uniform branch point the first-order term is exactly zero and the realized damage is two orders of magnitude below that of a peaked state under the same perturbation.

B Implementation Notes

Sibling mask. For a group of G rollouts of length T we form $\text{eq}[i, j, t] = \mathbf{1}[y_t^{(i)} = y_t^{(j)}]$ and take a cumulative product along t , so that $\text{prefixeq}[i, j, t]$ is true exactly when the two rollouts agree on $y_{\leq t}$. Shifting it one position right gives agreement on $y_{< t}$, which is the sibling relation Eq. (15) requires. Padded positions are treated as disagreement, so a length difference counts as a divergence.

Alignment. $\hat{H}_{\text{togo}}$ is read at position $t + 1$ and moved to position t ; at the final position the value is kept. Omitting this shift scores the state before the branch instead of the branch taken.

Memory. The RL path never materializes head logits. For each head the hidden states are processed in chunks of 4096 rows, mapped through the shared unembedding, reduced to an entropy, and released, so peak memory is $O(\text{chunk} \times V)$ rather than $O(\kappa \times B \times T \times V)$.

Equivalence test. A unit test asserts that with $\lambda = 0$ the token weights are bitwise identical to the base method’s on random inputs; the $\lambda = 0$ early return in the combination step is what makes this hold exactly rather than to floating-point tolerance.

C Pre-Registered Analysis

The analysis plan below was committed to the artifact repository before any run of the multi-seed campaign was launched, so that the statistics could not be chosen to fit the outcome. It is reproduced here in full; the file is `docs/preregistration_5seed.md` in the release.

Unit and primary endpoint. The unit of analysis is the seed, not the validation step. Each run is reduced to one number — its mean `acc@32` on

AIME24 over steps 40–110 — and arms are paired within a seed.

Contrasts, in the order they are reported. Primary: STEER-F – GRPO. Secondary, in this order: STEER-F – STEER, STEER-F – UNIFORM, STEER-F – PERMUTED. Secondary endpoints: `maj@32`, uplift, and the six-benchmark mean. Effect sizes are reported with intervals rather than significance stars, and all five arms are reported whatever the outcome.

Stopping and exclusion. Every run goes to step 110; none is stopped early for looking good or bad. A run is excluded only for mechanical failure — out of memory, a crash, a corrupted checkpoint, or a step-0 validation score outside .030–.055, which would indicate a different initial checkpoint. Exclusions are listed with their reason. A failed run is relaunched with the same seed, never a new one, and no seed is added after results are seen.

Checkpoint selection. Two rules are computed and both reported: the argmax over a run’s validation points, and a fixed step 110. The six-benchmark evaluation uses the argmax. Selecting a maximum over twelve noisy checkpoints is biased upward — in the single-seed data by $+.011$ to $+.017$ per arm — which is why the primary endpoint is a window mean rather than a maximum.

What would falsify the central claim. Stated in advance so it cannot be renegotiated afterwards. If STEER-F – UNIFORM straddles zero across seeds, the forecast’s *value* does not matter and the apparatus explains the effect; that is the claim of this paper and it would be refuted. If STEER-F – PERMUTED straddles zero, the pairing between a branch and its score does not matter, which the derivation says it must. If aggregate entropy orders the arms the same way accuracy does, the dissociation argument of Section 12.3 fails, since it depends on the two orderings coming apart. And if the mean token weight in the STEER-F arm departs from ≈ 0.999 , then A_H is not sum-zero over sibling sets as derived and the implementation is wrong rather than the theory.

Amendment, recorded. The pre-registration assigned three seeds to UNIFORM and PERMUTED and five to the other three arms. The campaign as executed runs all five arms at five seeds. The change raises the power of the two control contrasts and was made before any result of the campaign

was seen; it is recorded here rather than silently absorbed.

Second amendment, added 2026-09-21, after this section’s own entropy condition was retired.

Everything above was fixed before the campaign ran. This paragraph was not, and it is marked so rather than folded into the text above. It justifies nothing about the seeds already collected; it binds the seeds not yet in hand.

What it replaces. The third falsification bullet above put a condition on the aggregate entropy ordering. Section 12.3 retires it as a category error — it registers the aggregate as a signal about A_H , which Section 9.3 had already said it cannot be, an hour before this file was committed — and claims nothing from it in either direction. A first attempt at a replacement, registered earlier on 2026-09-21, named `steerf/branch_entropy`; that registration is withdrawn here rather than deleted, because the monitor it named selects a fixed top decile of A_H rather than the support (Section 12.6).

Registered now, before the runs that can answer it. The quantity is `steerf/support_entropy` and its gap, the entropy split on $A_H \neq 0$ — the same predicate the correction itself applies, so the split covers the positions the method touches and no others. The primary contrast is STEER-F – PERMUTED, paired by seed, because that pair shares λ , $\lambda_{\min}$, the sampler and the magnitudes of A_H , and differs only in whether a branch is paired with its own score; STEER-F – UNIFORM is secondary. The claim under test is that a support-restricted entropy contrast near zero can accompany an accuracy contrast away from zero.

What would falsify it. If the support-restricted contrast tracks the accuracy contrast in sign and magnitude across seeds the way the apparatus contrasts in Table 7 do, then the split is the aggregate statistic in a smaller window and the claim fails.

What is still not instrumented, and we say so rather than substitute. Section 9.3’s own prediction is that the spread *across siblings* widens. Testing it needs the sibling index at the loss site, where only the per-token tensors are currently in scope, so no quantity we log tests it today. We do not offer the support-restricted split as that test: it measures a level, not a spread.

D Within-Run Stability

Earlier drafts of this work paired the arms at each of the eight validation steps in the converged window and reported a paired t over those eight points. That statistic answers a narrower question than the one in Section 12.4: it asks whether a gap is stable along a single trajectory, not whether it reappears when the run is repeated. Its n counts validation points, which are serially correlated within a run, so it is not an estimate of run-to-run uncertainty and cannot be read as one.

We report it here because it is informative about a different thing — whether a gap is an artifact of one lucky validation point — and because the single-seed version of this manuscript quoted it. Where the two disagree, the seed-level statistic in Section 12.8 governs.

The contrast between the two is the point. Table 14 separates every A_H comparison at t_7 between +2.56 and +7.18, which reads as decisive and is not: the same gap sits at the same order as the run-to-run spread we measure in Section 12.8. A statistic can be this confident about a trajectory and say nothing about a re-run, which is exactly why we moved it out of the main text.

contrast	$\Delta\text{acc@32}$	paired t_7
STEER-F – GRPO	+0.0145	+3.08
STEER-F – STEER	+0.0165	+6.68
STEER-F – UNIFORM	+0.0182	+7.18
STEER-F – PERMUTED	+0.0125	+2.56

Table 14: Within-run stability over the eight converged-window validation points of seed 1. The points are consecutive validations of one run and are therefore serially correlated, so $\text{df} = 7$ is generous and these t values are not an estimate of run-to-run variation; see Section 12.8 for that. Majority-vote versions of the same four rows separate at +5.09 to +5.50.